DBC 7.8

Pressure-Modulation Electro-Hydraulic Control Unit

ECU Diagnostic Communication for ISO 14229

Customer Technical Document

Specification # 18105034

Rev 1.3

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Table of contents

1 INTRODUCTION 6

1.1 SCOPE OF DOCUMENT 6
1.2 MISSION/THEME 6
1.3 ORDER OF PRECEDENCE 6
1.4 APPLICABLE DOCUMENTS 6
1.5 REFERENCES 6

2 DBC 7.8 ECU FUNCTIONALITY 7

2.1 DIAGNOSTIC PROTOCOL 7
2.1.1 Physical Layer 7
2.1.2 Data Link Layer 7
2.1.2.1 Diagnostic CAN Identifiers 8
2.1.3 Network/transport layer 8
2.2 SELF TEST OPERATION 8
2.2.1 Initialization Self Test 8
2.2.2 Continuous Self Test 9
2.2.2.1 Continuous Self Test Cycle Time 9
2.2.2.2 Total Test Time 9
2.2.3 Fault Definition 9
2.2.3.1 DTC Number 9
2.2.3.2 DTC Description 9
2.2.4 General Diagnostic Fault Processing 10
2.2.4.1.1 Fault Detection/Test 10
2.2.4.1.2 Indication to Vehicle Operator 11
2.2.4.1.3 Clearing DTC History 11
2.2.4.1.4 DTC Information Storage 11
2.2.4.1.5 Failure Records and Enhanced Diagnostics 12
2.3 DIAGNOSTIC SERVICES 12
2.3.1 DBC 7.8 Supported Service Identifier value summary table 12
2.3.2 Diagnostic and Communication Management functional unit 14
2.3.2.1 DiagnosticSessionControl service (SId $10) 14
2.3.2.1.1 Request Message definition 14
2.3.2.1.2 Request Message Data Bytes 16
2.3.2.1.3 Positive Response Message Data Bytes 16
2.3.2.1.4 Negative Response Message Data Bytes 16
2.3.2.2 SecurityAccess service (SId $27) 17
2.3.2.2.1 Request Message definition 18
2.3.2.2.2 Positive Response Message 19
2.3.2.2.3 Negative Response Message 19
2.3.2.2.4 Implementation Note 20

1. REQUEST IS FUNCTIONAL ADDRESSED **20**

REQUESTSEQUENCEERROR 20

2.3.2.3 TesterPresent service (SId $3E) 20
2.3.2.3.1 Request Message definition 21
2.3.2.3.2 Request Message Data Bytes 21
2.3.2.3.3 Positive Response Message Data Bytes 21
2.3.2.3.4 Negative Response Message Data Bytes 21
2.3.3 Data Transmission functional unit 22
2.3.3.1 ReadDataByIdentifier service (SId $22) 22
2.3.3.1.1 Request Message definition 22
2.3.3.1.2 Positive Response Message Data Bytes 22
2.3.3.1.3 Negative Response Message Data Bytes 23

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

2.3.3.2 ReadMemoryByAddress service (SId $23) 23
2.3.3.2.1 Request Message definition 23
2.3.3.2.2 Request Message Data Bytes 24
2.3.3.2.3 Positive Response Message Data Bytes 24
2.3.3.2.4 Negative Response Message Data Bytes 24
2.3.3.3 WriteDataByIdentifier service (SId $2E) 25
2.3.3.3.1 Request Message definition 25
2.3.3.3.2 Request Message Data Bytes 25
2.3.3.3.3 Positive Response Message Data Bytes 25
2.3.3.3.4 Negative Response Message Data Bytes 25
2.3.3.4 Test service (SId $B0) 26
2.3.3.4.1 Request Message definition 26
2.3.3.4.2 Request Message Data Bytes 27
2.3.3.4.3 Positive Response Message 27
2.3.3.4.4 Negative Response Message 27
2.3.4 Stored Data Transmission functional unit 28
2.3.4.1 ClearDiagnosticInformation service (SId $14) 28
2.3.4.1.1 Request Message definition 28
2.3.4.1.2 Request Message Data Bytes 29
2.3.4.1.3 Positive Response Message Data Bytes 29
2.3.4.1.4 Negative Response Message Data Bytes 29
2.3.4.2 ReadDTCInformation service (SId $19) 30
2.3.4.2.1 Request Message definition 30
2.3.4.2.2 DTC status bit definitions 31
2.3.4.2.3 Request Message Data Bytes 31
2.3.4.2.4 Positive Response Message Data Bytes 31
2.3.4.2.5 Negative Response Message Data Bytes 32
2.3.5 InputOutput Control functional unit 32
2.3.5.1 InputOutputControlByIdentifier (Sid $2F) 33
2.3.5.1.1 Request Message definition 33
2.3.5.1.2 Request Message Data Bytes 33
2.3.5.1.3 Positive Response Message Data Bytes 34
2.3.5.1.4 Negative Response Message Data Bytes 34

3 SERVICE SCAN TOOL INTERFACE 35

3.1 SERVICE SCAN TOOL COMMUNICATIONS WITH CONTROLLER 35
3.2 DIAGNOSTIC TROUBLE CODE INFORMATION 35
3.2.1 Read Current Diagnostics Codes 35
3.2.2 Read Enhanced Diagnostic History 35
3.2.3 Clear Diagnostic Codes 35
3.3 ECU IDENTIFICATION 36
3.4 DATA DISPLAY FUNCTIONS 36
3.5 INPUT / OUTPUT DEVICE CONTROL 37
3.5.1 Hydraulic Output Control 38
3.5.1.1 Solenoid Control 38
3.5.1.2 Pump Control 38
3.5.1.3 Brake System Bleed 38

4 PARAMETER DEFINITIONS 39

4.1 PARAMETER CODING 39
4.2 BIT AND BYTE ORDER 39
4.3 DATAIDENTIFIER DEFINITION 39
4.3.1 Static DataIdentifier 40
4.3.1.1 DID $0100 OEMSpecific_1 (Read/Write) 40
4.3.1.2 DID $F182 applicationDataIdentificationDataIdentifier (Read/Write) 40
4.3.1.3 DID $F183 ECUBootloaderSoftwareReferenceNumber (Read) 41
4.3.1.4 DID $F187 ECUPartNumber (Read/Write) 41
** Value will be HEX, USN, or ASCII as required by application. 41
4.3.1.5 DID $F18A SystemSupplierIdentifier(Read) 41
4.3.1.6 DID $F18B ECUManufactureDate(Read) 41
4.3.1.7 DID $F190 vehicleIdentificationNumber (Read/Write) 42
4.3.1.8 DID $F192 systemSupplierECUHardwareNumber (Read) 42

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

4.3.1.9 DID $F193 systemSupplierECUHardwareVersionNumber (Read) 42
4.3.1.10 DID $F194 systemSupplierECUSoftwareNumber (Read) 42
4.3.1.11 DID $F195 SystemSupplierECUSoftwareVersionNumberDataIdentifier(Read) 42
4.3.1.12 DID $F197 SystemName(Read) 43
4.3.1.13 DID $F199 Date ECU Reprogrammed (Read/ Write*) 433
4.3.1.14 DID $F19B CalibrationDateInformation (Read/Write) 43
4.3.1.15 DID $F19D ECUInstallationDateDataIdentifier (Read/ Write) 43
4.3.1.16 DID $F1A0 OEMSpecific_2(Read/Write) 44
4.3.1.17 DID $F1A1 OEMSpecific_3(Read/Write) 444
4.3.1.18 DID $3003 SupplierEHCUTraceability (Read/Write) 44
4.3.1.19 DID $3020 ManufacturersEnableCounter (Read/Write) 44
4.3.2 Supplier Defined Dynamic DataIdentifier 45
4.3.2.1 DID $3007 System Speed and Status Information 45
4.3.2.2 DID $3008 Power, Motor and Solenoid Information 46
4.3.2.3 DID $3009 Current And History ECU MCU DTC Information 47
4.3.2.4 DID $300A Last Fault Diagnostic Information 47
4.3.2.5 DID $300B Second Last Fault Diagnostic Information 49
4.3.2.6 DID $300C Third Last Fault Diagnostic Information 50
4.3.2.7 DID $3017 Fourth Last Fault Diagnostic Information 51
4.3.2.8 DID $3018 Fifth Last Fault Diagnostic Information 53
4.3.2.9 DID $300D CurrentDiagnosticDTCInformation 54
4.3.2.10 DID $300E HistoryDiagnosticDTCInformation 54
4.3.3 DID $3100 ~ $311F DeviceControl 55
4.3.3.1 DataIdentifier #1 Definition 55
4.3.3.2 Control State bytes definition 55

5 REVISION HISTORY 57

List of Tables in the Document:

Table 2.3-1 DBC 7.8 Supported Service Identification Value Summary Table 13
Table 2.3-2 Diagnostic and Diagnostic Management functional unit 14
Table 2.3-3 Definition of diagnosticSession Values 14
Table 2.3-4 DBC 7.8 Supported Default Service Identification Value Summary Table 15
Table 2.3-5 ExtendedDiagnsoticSession Service Identification Value Summary Table 15
Table 2.3-6 SafetySystemDiagnosticSession Service Identification Value Summary Table 16
Table 2.3-7 DiagnosticSessionControl Request Message 16
Table 2.3-8 DiagnosticSessionControl Positive Response Message 16
Table 2.3-9 Diagnostic Services $10 Response Code 17
Table 2.3-10 TesterPresent Request Message 21
Table 2.3-11 TesterPresent Positive Response Message 21
Table 2.3-12 TesterPresent Negative Response Message 21
Table 2.3-13 Diagnostic Services $3E Response Code 21
Table 2.3-14 ReadDataByIdentifier Request Message 22
Table 2.3-15 ReadEcuIdentification Positive Response Message 23
Table 2.3-16 ReadDataByIdentifier Negative Response Message 23
Table 2.3-17 Diagnostic Services $21 Response Code 23
Table 2.3-18 ReadMemoryByAddress Request Message 24
Table 2.3-19 ReadMemoryByAddress Positive Response Message 24
Table 2.3-20 ReadMemoryByAddress Negative Response Message 24
Table 2.3-21 Diagnostic Services $23 Response Code 24
Table 2.3-22 — Request message data parameter definition 25
Table 2.3-23 WriteDataByIdentifier Request Message 25
Table 2.3-24 WriteDataByIdentifier Positive Response Message 25
Table 2.3-25 WriteDataByIdentifier Negative Response Message 26
Table 2.3-26 Diagnostic Services $2E Response Code 26
Table 2.3-27 Request message sub function definition 27
Table 2.3-28 TestService Request Message 27
Table 2.3-29 TestService Positive Response Message 27
Table 2.3-30 Testservice Negative Response Message 27
Table 2.3-31 Diagnostic Services $B0 Response Code 28
Table 2.3-32 Stored Data Transmission functional unit 28
Table 2.3-33 — Request message data parameter definition 28

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Table 2.3-34 ClearDiagnosticInformation Request Message 29
Table 2.3-35 ClearDiagnosticInformation Positive Response Message 29
Table 2.3-36 ClearDiagnosticInformation Negative Response Message 29
Table 2.3-37 Diagnostic Services $14 Response Code 30
Table 2.3-38 Request message sub function definition 30
Table 2.3-39 — Request data parameter definition 30
Table 2.3-40 Definition of statusOfDTC values 31
Table 2.3-41 ReadDiagnosticTroubleCodesByStatus Request Message 31
Table 2.3-42 ReadDiagnosticTroubleCodesByStatus Positive Response Message 31
Table 2.3-43 — Response message definition - sub-function = reportDTCByStatusMask, reportSupportedDTCs 32
Table 2.3-44 ReadDiagnosticTroubleCodesByStatus Negative Response Message 32
Table 2.3-45 Diagnostic Services $18 Response Code 32
Table 2.3-46 InputOutput Control functional unit 32
Table 2.3-47 Request message data parameter definition 33
Table 2.3-48 InputOutputControlByIdentifier request message 34
Table 2.3-49 InputOutputControlByIdentifier positive response message 34
Table 2.3-50 InputOutputControlByIdentifier negative response message 34
Table 4.3-1 DataIdentification Parameter definition 40
Table 4.3-2 DID $3100-$31FF System Output Control 56

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

1 Introduction

This document is intended as a reference document, which describes the diagnostic functionality and diagnostic communication services of the DBC 7.8 ECU. This document is **NOT** the specification for the diagnostic functionality of the DBC 7.8 ECU.

1.1 Scope of Document

This document describes the technical details with reference to on-board assembly line and service diagnostics for the ISO 14229 DBC 7.8 applications. It is to be used in conjunction with following documentation provided for the DBC78 PM EHCU Brake Control

DBC 7.8 - Pressure-Modulation Electro-Hydraulic Control Unit, Use and Application
Customer Technical Document

DBC 7.8 - Pressure-Modulation Electro-Hydraulic Control Unit, Diagnostic and Service Manual

DBC78 Customer Diagnostic Code Cross Reference Table

DBC78 Customer Mechanization Drawing

1.2 Mission/Theme

The DBC 7.8 ECU is equipped with integrated self diagnostic functionality capable of determining failures in the EHCU and its external inputs (i.e. sensors, etc.). The ECU uses the self diagnostic functionality to prevent damage to components and to ensure that the ECU behaves in a predictable and safe manner in the presence of malfunctions. The ECU is equipped with additional capabilities to support ECU and EHCU manufacturing, vehicle assembly activities as well as testing in a vehicle assembly or service environment.

1.3 Order of Precedence

In the event of a conflict between the text of this document and the documents cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

This document is **NOT** the controlling specification for the features and functionality described within.

1.4 Applicable Documents

See section 11 of "DBC 7.8 - Pressure-Modulation Electro-Hydraulic Control Unit, Use and Application Customer Technical Document", Specificiation number, 18083939.

1.5 References

See section 12 of "DBC 7.8 - Pressure-Modulation Electro-Hydraulic Control Unit, Use and Application Customer Technical Document", Specificiation number, 18083939.

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

2 DBC 7.8 ECU Functionality

2.1 Diagnostic Protocol

2.1.1 Physical Layer

The High Speed CAN bus physical layer conforms to the requirements defined in ISO 11898-2:Road vehicles — Controller area network (CAN) — Part 2: High-speed medium access unit. Refer to that document for details.

2.1.2 Data Link Layer

According to ISO 11898-1: Road vehicles — Controller area network (CAN) — Part 1: Data link layer and physical signaling.

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

2.1.2.1 Diagnostic CAN Identifiers

Diagnostic CAN identifiers shall be implemented according to the vehicle communication requirements for the customer vehicle application.

2.1.3 Network/transport layer

According to ISO 15765-2:Road vehicles — Diagnostics on controller area network (CAN) — Part 2: Network layer service.

2.2 Self Test Operation

2.2.1 Initialization Self Test

The ECU initialization self tests will be executed in the DBC78 PM ECU immediately after the ignition voltage input state transitions from low to high. The following will be tested during the ECU Initialization Self Tests:

- RAM. Verify each cell. Each cell of used RAM is verified using read/write tests with defined bit patterns.
- EEPROM. All system calibrations specific EEPROM will be checksummed as part of initialization. This checksum includes system calibrations such as sensor trim data.
- Flash EPROM. All program flash will be checksummed as part of initialization.
- A/D Inputs. Verify correct reading of VRH, VRL, and (VRH + VRL)/2.
- Wheel Speed Circuit hardware integrity

The DBC78 PM ECU will perform a modulator initialization self test by energizing its hydraulic actuators. This test is executed under conditions which will minimize any disturbance to the driver; The test will be executed when the following set of conditions are satisfied for the first time in an ignition cycle:

1. System brake is not applied
2. Engine RPM has reached a level that insures the engine is running (this criteria is used only on applications that have engine RPM data available through high speed serial data).
3. ABS or DRP is not active
4. Vehicle speed is less than a low speed threshold

If the conditions above are not present prior to the vehicle achieving a minimum speed (typically 16 kph), the initialization self test will be executed. The initialization self test is not executed when the vehicle is operated in the Device Control mode.

This initialization period consists of commanding the motor on and commanding each of the solenoids on. Initialization of the motor and solenoids is performed in order to detect possible “On State” hardware faults at the beginning of every drive cycle and before the driver enters an actual ABS or DRP event. “On State” hardware faults are faults that can only be detected when the hardware is commanded on.

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

2.2.2 Continuous Self Test

The following tests will be carried out during the continuous self test. If a detected fault could be regarded as a potential hazard, actions will be taken to ensure safe system behavior.

- RAM. Verify read/write operations.
- Flash EPROM. All program Flash will checksummed as a background task during application execution.
- EEPROM. All run-time accessed EEPROM (i.e. calibration data) will be checksummed as a background task during application execution.
- A/D Inputs. Verifies that separate A/D readings of the same voltage correlate.
- Analog Sensors. Check voltage limits and for unstable readings. When applicable information from multiple sensors will be checked for correlation.
- Switches. When applicable checks for open and invalid switch combinations will be performed.
- Electromechanical actuators loads - Checks for open/shorts will be performed

2.2.2.1 Continuous Self Test Cycle Time

- RAM. Typical cycle time does not exceed 5 seconds (actual cycle time varies with loop processing load) .
- Flash EPROM. Typical cycle time does not exceed 5 seconds (actual cycle time varies with loop processing load).
- EEPROM. Typical cycle time does not exceed 5 seconds (actual cycle time varies with loop processing load).
- A/D Inputs. Once per software execution loop.
- Analog Sensors. The cycle time for any given sensor is dependent on the enable criteria and calibration settings; Refer to DBC78 PM EHCU Diagnostic and Service Manual for additional details.
- Switches. The cycle time for any given switch is dependent on the enable criteria and calibration settings; Refer to DBC78 PM EHCU Diagnostic and Service Manual for additional details.
- Actuators, loads. The cycle time for any given actuator/load is dependent on the enable criteria and calibration settings; Refer to DBC78 PM EHCU Diagnostic and Service Manual for additional details.

2.2.2.2 Total Test Time

The total test time for initialization and a complete run of all self tests is dependent on meeting the enable criteria for all of the tests to runDiagnostic Fault Detection and DTC setting

2.2.3 Fault Definition

A fault is a malfunction, which will cause noticeable abnormal system behavior, performance problems or damage. Additional system faults which may not have a noticeable, adverse effect can be detected in support of customer safety or compliance to governmental safety standards.

2.2.3.1 DTC Number

The numbering of Diagnostic Trouble Codes (DTC) will be made per the C0XXX standard for the DBC78 ECU. Refer to the Customer Application Diagnostic Code Cross Reference. This document will list the following:

1. The applicable diagnostic tests for a customer application vehicle.
2. The corresponding DBC78 Fault Code Number in C0XXX format
3. Any customer specific fault code reporting formats that are used for diagnostic communications.

2.2.3.2 DTC Description

The descriptions of the DTC are defined in the document “DBC78 PM ECHU Diagnostic and Service Manual”. The document includes detailed descriptions of:

1. Failure type definitions
2. Enable criteria including vehicle power modes, ECU operating modes
3. Secondary fault setting criteria and exceptions.
4. Fault Actions.

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

2.2.4 General Diagnostic Fault Processing

The following paragraphs outline the basic process for all diagnostic trouble code processing.

2.2.4.1.1 Fault Detection/Test

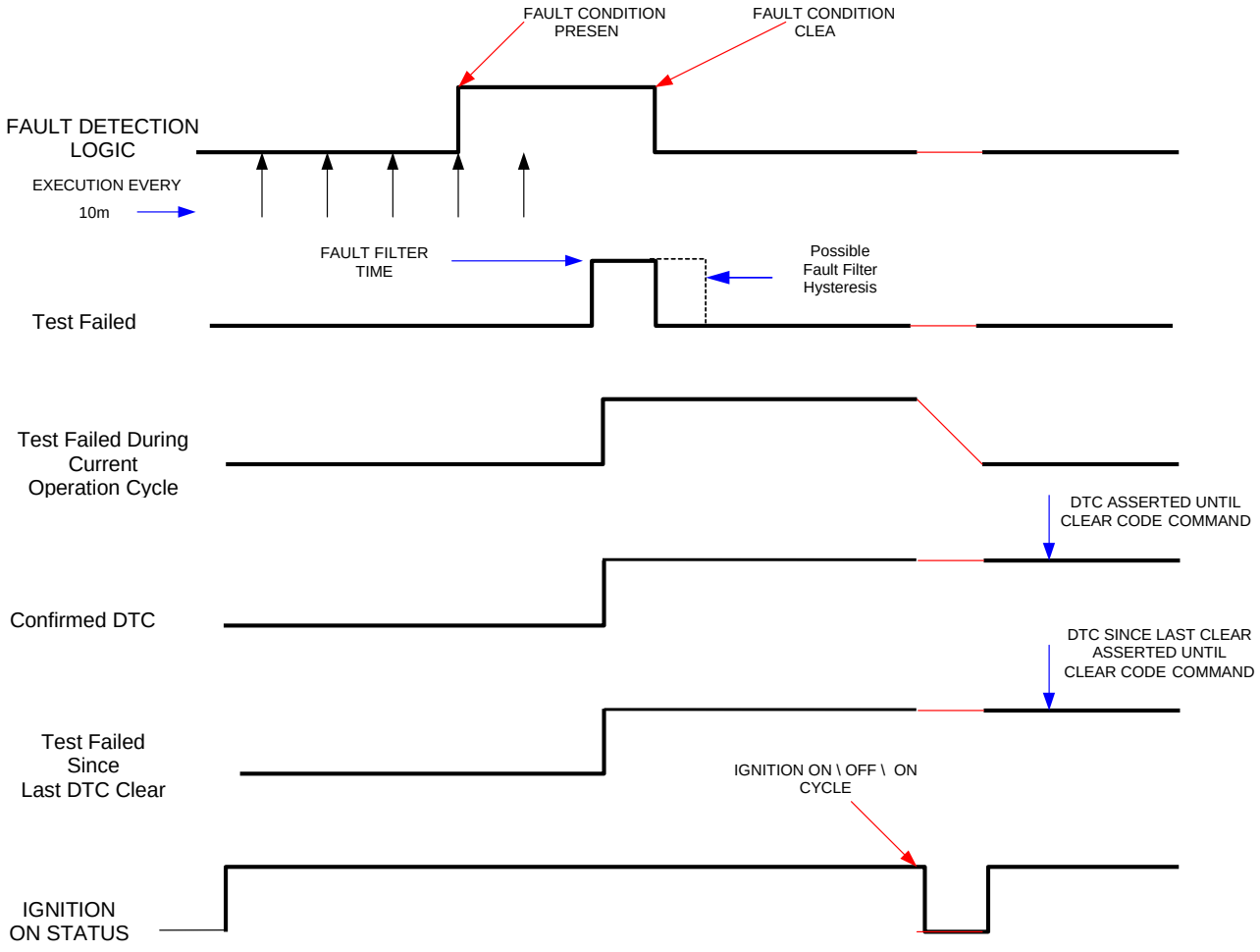
A diagnostic test performed by the DBC78 EHCU is defined as the following:

Actions performed from a given moment when all conditions to start a diagnostic fault detection algorithm are fulfilled until the moment a diagnostic fault detection algorithm makes a pass or fail decision.

A fault should only be detected when a real fault exists. In order to prevent false failure conditions, all necessary logic (e.g. necessary filtering) to determine whether a real fault exists will be performed before a result is determined.

Secondary failures are generally not logged. Wherever possible the diagnostic fault detection algorithms will be designed such that a single point failure results in a single failure being logged. Exceptions will be documented in the "DBC78 Diagnostics and Service Manual".

When a diagnostic fault test asserts a DTC due to a failure detected in the current ignition cycle, the DTC will be stored to NVM (Non-Volatile Memory) when the DBC78 EHCU is powered down. Also, during EHCU power down, the DTC will be stored with a status of "history". The DTC will then be reported with a status of "history" on subsequent ignition cycles.
The figure below illustrates the processing of a DTC that occurs during an ignition cycle:



DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

2.2.4.1.2 Indication to Vehicle Operator

If information concerning a fault is judged to be important for the vehicle operator, the fault information will typically be indicated by illuminating a warning lamp. Specific fault indication actions are described in the “DBC78 Diagnostic and Service Manual”.

The vehicle operator indication will remain present for as long as the detected fault condition is present. DTCs that do not have an active fault condition, but are stored with a status of “history” will not generate a fault indication to the vehicle operator.

2.2.4.1.3 Clearing DTC History

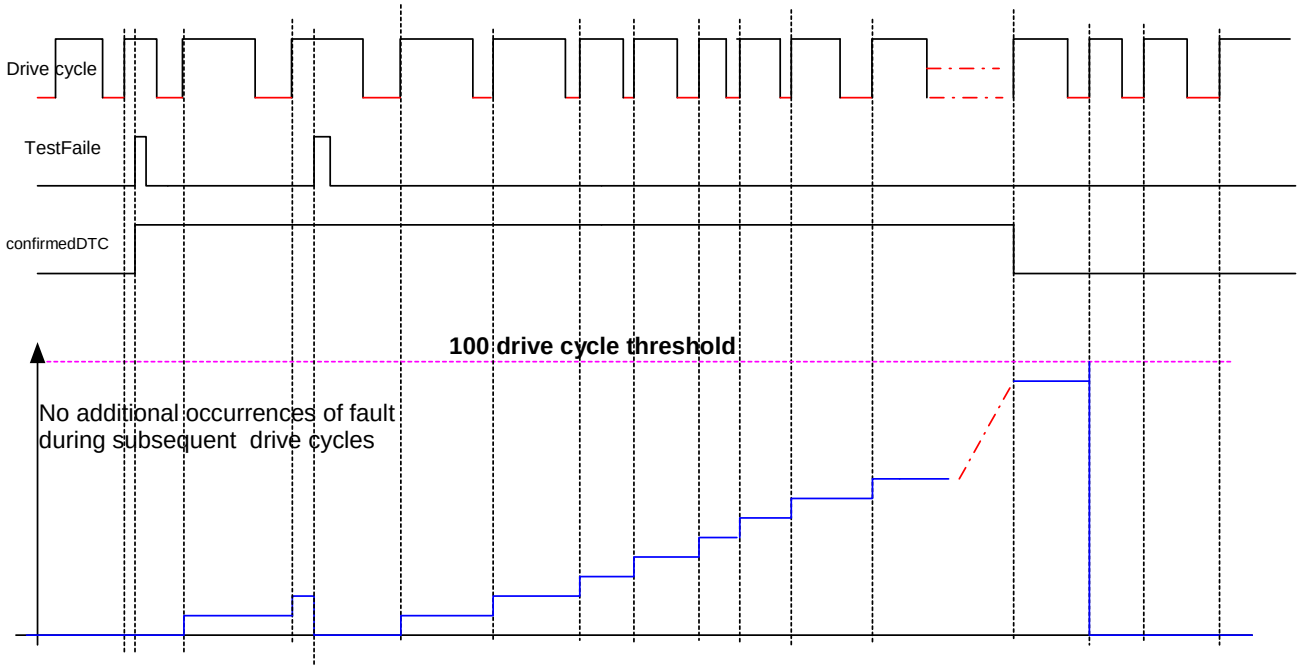
The deletion of a DTC with a status of “history” will be possible with the diagnostic service ”ClearDiagnosticInformation”.

In addition the DBC7.8 EHCU will implement an ”Automatic Fault Recovery” function.

In general, the Automatic Fault Recovery function will operate as follows:

The fault information for a specific diagnostic test will be cleared after 100 consecutive drive cycles with no assertions of any other Fault Test conditions. If a Failure Record has been stored, the content of the failure record will be cleared and made available for a new failure record storage by any Fault Test. A drive cycle occurs after vehicle speed exceeds a minium threshold (typical value is 16 kph) during the power ON period, and the ECU transitions from the ON to OFF state

The figure below illustrates the Automatic Fault Recovery function:



2.2.4.1.4 DTC Information Storage

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

The fault test failures will be kept in non-volatile memory. When a request codes $18 service is received by the ECU, the recorded fault test failures will be mapped to a DTC Number. The DTC number and associated information will then be reported to the diagnostic test tool.

The DTC EEPROM information is stored to EEPROM when ignition is removed from the DBC 7.8 EHCU. If battery is lost during an ignition cycle in which faults were recorded, the fault information for that ignition cycle will not be stored to EEPROM.

2.2.4.1.5 Failure Records and Enhanced Diagnostics

The DBC78 PM EHCU has the capability to store five failure records in non volatile memory (EEPROM).
The failure records will include various system and vehicle conditions at the time when a fault code occurred, when last fault last occurred, the second last fault last occurred, and so on, up to five faults.
This feature is known as enhanced diagnostics. The content of the enhanced diagnostic failure records is described in the Section 4.3.2.5 through 4.3.2.8.

When a DTC that does not have an existing failure record is recorded, the failure record information at the time of DTC setting will be captured. If the DTC already has a failure record from a previous ignition cycle, then only the “Ignition Cycles Since Last Fault Code Set” and “Number of Occurrences of DTC” will be updated. All other data will reflect the data values from the first time the code was set.

The last five DTCs which support failure records that occur in an ignition cycle will have failure records stored. If additionalDTCs are asserted during the ignition cycle, only the 5 most rececent DTCs will have failure records stored. A new failure record will be placed into an empty failure record, if available. If all failure record areas are populated with data from previous ignition cycles when a new failure is recorded, the new DTC and its failure record information will replace the oldest failure record’s information.

Executing a clear codes diagnostic message operation will clear the failure record information for any DTC event, including all its stored failure record data. The newly cleared failure record area will be made available for data storage by any new DTC.

2.3 Diagnostic Services

Diagnostic Services will be implemented mainly according to ISO 14229 – 1; Additional details are described in this section.

2.3.1 DBC 7.8 Supported Service Identifier value summary table

The purpose of the "DBC 7.8 Supported Service Identifier value summary table" is to provide an overview about all services specified/referenced in this document. Any other attempted service request would be responded with serviceNotSupported (SNS, $11).

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Diagnostic Services	Req. Value
DiagnosticSessionControl	10
ClearDiagnosticInformation	14
ReadDTCInformation	19
ReadDataByIdentifier	22
ReadMemoryByAddress	23
Security Access	27
WriteDataByIdentifier	2E
InputOutputControlByIdentifier	2F
TesterPresent	3E
TestService	B0

Table 2.3-1 DBC 7.8 Supported Service Identification Value Summary Table

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

2.3.2 Diagnostic and Communication Management functional unit

In this unit DBC 7.8 supports the diagnostic services listed in the following table,

Service	Description
DiagnosticSessionControl	The client requests to control a diagnostic session with a server(s).
SecurityAccess	The client requests to unlock a secured server(s).
TesterPresent	The client indicates to the server(s) that it is still present.

Table 2.3-2 Diagnostic and Diagnostic Management functional unit

2.3.2.1 DiagnosticSessionControl service (SId $10)

This service is used to enable different diagnostic Sessions in the server. DBC 7.8 uses this service to enter safetySystemDiagnosticSession (DVT mode)，ExtendedDiagnsoticSession or go back to the defaultSession.

When the system is in the safetySystemDiagnosticSession.

1. Most of the system control functions like ABS are disabled.
2. DRP will remain enabled unless the I/O Control Service InputOutputControlByIdentifier (Sid $2F)is active
3. The appropriate system status signals / commands indicating the DRP, ABS functional statuses will be sent
4. Normal diagnostic (DTC) algorithms will be run until the I/O Control Service InputOutputControlByIdentifier (Sid $2F) is active. This will guarantee that all DTC data is accurate when the test tool checks for DTC's prior to the first I/O Control Service.
5. InputOutputControlByIdentifier ($2F) may be entered at any vehicle speed.

2.3.2.1.1 Request Message definition

2.3.2.1.1.1 sub-function Parameter Level definition

The sub-function parameter diagnosticSessionType is used by the DiagnosticSessionControl service to select the specific behaviour of the server. Explanations and usage of the possible diagnostic sessions are detailed below. The following sub-function values are specified (suppressPosRspMsgIndicationBit (bit 7) not shown)

Hex	Description	Mnemonic
01	**defaultSession** This diagnostic session enables the default diagnostic session in the server(s) and does not support any diagnostic application timeout handling provisions	DS
03	**extendedDiagnosticSession** This diagnosticSession can e.g. be used to enable all diagnostic services required to support the adjustment of functions. It can also be used to enable diagnostic services, which are not specifically tied to the adjustment of functions.	EXTDS
04	**safetySystemDiagnosticSession** This diagnosticSession enables all diagnostic services required to support safety system related functions e.g. airbag deployment.	SSDS

Table 2.3-3 Definition of diagnosticSession Values

Diagnostic session transition description:

- When the server transitions from diagnostic session to diagnostic session including the currently active diagnostic session, the server shall re-initialize completely. The server shall reset all activated/initiated/changed settings/controls during the activated session. This does not include long term changes programmed into non-volatile memory.

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

In all but the default diagnostic session the tester present timer is activated Any request of services listed in Table 2.3-1 DBC 7.8 Supported Service Identification Value Summary Table shall be responded with a negative code $22 if it is not available under currently active diagnostic session.

2.3.2.1.1.1.1 DefaultSession

If no diagnostic session has been requested by the client after system initialization default session shall be automatically enabled in the server. The default session shall support the following services.

Diagnostic Services	Default Session Service
DiagnosticSessionControl	10
ClearDiagnosticInformation	14
ReadDTCInformation	19
ReadDataByIdentifier	22
ReadMemoryByAddress	23
SecurityAccess	27
WriteDataByIdentifier	2E
TesterPresent	3E
Test	B0

Table 2.3-4 DBC 7.8 Supported Default Service Identification Value Summary Table

2.3.2.1.1.1.2 ExtendedDiagnosticSession

Upon this diagnostic session actives, the DBC 7.8 shall be in the ECU extended diagnostic Session and the following diagnostic services shall be enabled.

Diagnostic Services	Req. Value
DiagnosticSessionControl	10
ClearDiagnosticInformation	14
ReadDTCInformation	19
ReadDataByIdentifier	22
ReadMemoryByAddress	23
SecurityAccess	27
WriteDataByIdentifier	2E
InputOutputControlByIdentifier	2F
TesterPresent	3E
Test	B0

Table 2.3-5 ExtendedDiagnsoticSession Service Identification Value Summary Table

2.3.2.1.1.1.3 SafetySystemDiagnosticSession

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Upon this diagnostic Session active, the DBC 7.8 shall be in the DVT mode and the following diagnostic services shall be enabled.

Diagnostic Services	Req. Value
DiagnosticSessionControl	10
ReadDTCInformation	19
ReadDataByIdentifier	22
ReadMemoryByAddress	23
InputOutputControlByIdentifier	2F
TesterPresent	3E

Table 2.3-6 SafetySystemDiagnosticSession Service Identification Value Summary Table

2.3.2.1.1.2 Data parameter definition

This service does not support data parameters in the request message.

2.3.2.1.2 Request Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	DiagnosticSessionContro Request Service Id	M	10	DSC
#2	sub-function = [defaultSession extendedDiagnosticSession safetySystemDiagnosticSession]	M	 01 03 04	LEV- DS EXTDS- SSDS

Table 2.3-7 DiagnosticSessionControl Request Message

2.3.2.1.3 Positive Response Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	DiagnosticSessionContro Positive Response Service Id	S	50	STDSPR
#2	DiagnosticSession= [defaultSession extendedDiagnosticSession safetySystemDiagnosticSession]	M	 01 03 04	DCM DS EXTDS- SSDS

Table 2.3-8 DiagnosticSessionControl Positive Response Message

2.3.2.1.4 Negative Response Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	DiagnosticSessionContro Request Service Id	S	10	STDS
#3	ResponseCode = [subFunctionNotSupported incorrectMessageLengthOrInvalidFormat ConditionsNotCorrect]	M	 12 13 22	 SFNS IMLOIF CNC

Figure 2.3-1 DiagnosticSessionControl Negative Response Message

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Hex	Conditions
12	subFunctionNotSupported Requested DiagnosticSession is not supported
13	incorrectMessageLengthOrInvalidFormat 1. Wrong length for the specific request
22	ConditionsNotCorrect 1. This response code will occur if the node (ECU) internal conditions do not allow the change of diagnostic Session. 2. The system is currently in initialization process. The system is in the process of changing EEPROM. 3. The system has fault which causing the system relay to be turning off.

Table 2.3-9 Diagnostic Services $10 Response Code

2.3.2.2 SecurityAccess service (SId $27)

The purpose of this service is to provide a means to access data and/or diagnostic services, which have restricted access for security, emissions, or safety reasons.
A typical example of the use of this service is as follows:

- client requests the “Seed”
- server sends the “Seed”
- client sends the “Key” (appropriate for the Seed received)
- server responds that the “Key” was valid and that it will unlock itself

A vehicle manufacturer specific time delay (10 seconds) shall be required before the server can positively respond to a service SecurityAccess ‘requestSeed’ message from the client after server power up/reset and after a certain number of false access attempts (see further description below). The delay shall be activated after a vehicle manufacturer specified number of false access attempts has been reached or when the server is powered up/reset and a previously performed SecurityAccess service has failed due to a single false access attempt. After a successful SecurityAccess service 'sendKey' execution the server internal indication information for a delay timer invocation (we only support one pare of security seed and key for DID write protection at this time) on a power up/reset shall be set to FALSE by the server. In case the server cannot determine if a previously performed SecurityAccess service prior to the power up/reset has failed then the delay timer shall always be active after power up/reset. The delay is only required if the server is locked when powered up/reset.

The client shall request the server to "unlock" by sending the service SecurityAccess ‘requestSeed’ message. The server shall respond by sending a "seed" using the service SecurityAccess ‘requestSeed’ positive response message. The client shall then respond by returning a "key" number back to the server using the appropriate service SecurityAccess ‘sendKey’ request message. The server shall compare this "key" to one internally stored. If the two numbers match, then the server shall enable ("unlock") the client's access to specific services/data and indicate that with the service SecurityAccess ‘sendKey’ positive response message. If the two numbers do not match, this shall be considered a false access attempt. If access is rejected for any other reason, it shall not be considered a false access attempt. An invalid key requires the client to start over from the beginning with a SecurityAccess 'requestSeed' message.

If a server is already unlocked when a SecurityAccess ‘requestSeed’ message is received, that server shall respond with a SecurityAccess ‘requestSeed’ positive response message service with a seed value equal to zero (0). The client shall use this method to determine if a server is locked by checking for a non-zero seed.

There shall always be a fixed relationship for each level of security supported so that the sendKey sub-function parameter value used for any given security level shall be equal to the requestSeed sub-function parameter value used for that security level plus one.

Only one security level shall be active at any instant of time. For example, if the security level associated with requestSeed 03 hex is active and a tester request is successful in unlocking the security level associated with requestSeed 01 hex, then only the secured functionality supported by the security level associated with requestSeed 01 hex shall be unlocked at that time. Any additional secured functionality that was previously unlocked by the security level associated with requestSeed 03 hex shall no longer be active. The security levels numbering is arbitrary and does not imply any relationship between the levels.

Attempts to access security shall not prevent normal vehicle communications or other diagnostic communication.
Servers, which provide security shall support reject messages if a secure service is requested while the server is locked.

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Some diagnostic functions/services requested during a specific diagnostic session may require a successful security access sequence. In such case the following sequence of services shall be required:

- DiagnosticSessionControl service
- SecurityAccess service
- Secured diagnostic service

There are different accessModes allowed for an enabled diagnosticSession (session started) in the server.

IMPORTANT — The server and the client shall meet the request and response message behaviour as specified in ISO 14229-1 section 7.5.2 Request message with sub-function parameter and server response behaviour in the event that those addressing methods are implemented for this service.

Please note that some customer applications will only use one set of seed and key pair for both DID write and reprogramming securities. Others will use each set for one single purpose.

This service is only supported if requested by in Physical addressing mode.

2.3.2.2.1 Request Message definition

2.3.2.2.1.1 sub-function Parameter Level definition

The sub-function parameter securityAccessType indicates to the server the step in the process for this service, the level of security the client wants to access and the format of the seed and key.

Examples:

- “requestSeed=01 hex” identifies a fixed relationship between “requestSeed=01 hex” and “sendKey=02 hex”
- “requestSeed=03 hex” identifies a fixed relationship between “requestSeed=03 hex” and “sendKey=04 hex”

Values are defined in 2.2.4.1.5-1 for requestSeed and sendKey (suppressPosRspMsgIndicationBit (bit 7) not shown).

Hex (bit 6-0)	Description	Cvt	Mnemonic
61	**requestSeed** Request DID write protection Seed	U	RSD
62	**Send Key** Send DID write protection Key	U	SK
01	**requestSeed** Request programming Seed Request DID write protection Seed	U	RSD
02	**Send Key** Send Programming Key Send DID write protection Key	U	SK

Table 2.3-10 Request message sub-function parameter definition

2.3.2.2.1.2 data parameter definition

This service does not support data parameters in the request message.

2.3.2.2.1.3 Request Message Data Bytes

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	SecurityAcces Request Service Id	M	27	SA
#2	sub-function = [securityAccessType = requestSeed]	M	01,61	LEV_ SAT_RSD

Table 2.3-11 Request message definition - sub-function = requestSeed

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	SecurityAcces Request Service Id	M	27	SA
#2	sub-function = [securityAccessType = sendKey]	M	02, 62	LEV_ SAT_SK
#3 #4	securityKey[] = [key#1 (high byte) key#2 (low byte)]	M M	00-FF 00-FF	SECKEY_ KEYHB KEYLB

Table 2.3-12 Request message definition - sub-function = sendKey

2.3.2.2.2 Positive Response Message

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	**SecurityAccess Response Service Id**	**S**	**67**	**SAPR**
#2	securityAccessType	M	01, 61	SAT_
#3 #4	securitySeed[] = [seed#1 (highbyte) seed#m (low byte)]	C C	00-FF 00-FF	SECSEED_ SEED1HB SEED2LB

Table 2.3-13 SecurityAccess requestSeed Positive Response Message

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	**SecurityAccess Response Service Id**	**S**	**67**	**SAPR**
#2	securityAccessType	M	02,62	SAT_

Table 2.3-14 SecurityAccess sendKey Positive Response Message

2.3.2.2.3 Negative Response Message

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	**negativeResponse Service Id**	**S**	**7F**	**NR**
#2	**testerPresent Request Service Id**	**M**	**27**	**TP**
#3	ResponseCode = [subFunctionNotSupported incorrectMessageLengthOrInvalidFormat ConditionsNotCorrect requestSequenceError invalidKey exceededNumberOfAttempts requiredTimeDelayNotExpired serviceNotSupportedInActiveSession]	M	 12 13 22 24 35 36 37 7F	**RC** SFNS IMLOIF CNC RSE IK ENOA RTDNE SNSIAS

Table 2.3-15 testerPresent Negative Response Message

2.3.2.2.4 Implementation Note

ResponseCode	Condition
12	**subFunctionNotSupported** 1. Requested sub-function is not supported
13	**incorrectMessageLengthOrInvalidFormat** 1. If COMPILE_MIN_LENGTH_CHECK_ONLY = FALSE Wrong length for the specific request 2. If COMPILE_MIN_LENGTH_CHECK_ONLY = TRUE length shorter than specified
22	**ConditionsNotCorrect** 1. Request is functional addressed
24	**requestSequenceError** 1. Send if the 'sendKey' sub-function is received without first receiving a 'requestSeed' request message.
35	**invalidKey** 1. Send if an expected 'sendKey' sub-function value is received and the value of the key does not match the server's internally stored/calculated key.
36	**exceededNumberOfAttempts** 1. Send if the delay timer is active due to exceeding the maximum number of allowed false access attempts.
37	**requiredTimeDelayNotExpired** 1. Send if the delay timer is active and a request is transmitted.
7F	**serviceNotSupportedInActiveSession** 1. The service is not enabled in the current diagnostic session.

Table 2.3-16 Diagnostic Services $27 Response Code

Manufacture Enable Counter and its value:
A manufacturers enable counter is used to bypass the security check during manufacturing.

2.3.2.3 TesterPresent service (SId $3E)

This service shall be used to indicate to a server that the client is present. This service is required in the absence of other service requests to prevent servers from automatically returning to default diagnostic session.. With each received

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

TesterPresent service or any other services that send to DBC 7.8, the DBC 7.8 shall reset its $S3_{server}$ timer. The presence of this service shall indicate that the system should remain in a diagnostic Session of operation.

2.3.2.3.1 Request Message definition

There are no parameters required in this diagnostic service.

2.3.2.3.1.1 Sub-function Parameter Level definition

Support 00/80 sub-function. Byte value “00” can receive positive response and value “80” suppress Positive Response.

2.3.2.3.1.2 Data parameter definition

This service does not support data parameters in the request message.

2.3.2.3.2 Request Message Data Bytes

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	testerPresent Request Service Id	M	3E	TP
#2	Response Required = [zeroSubFunction]	M	00/80	LEV_ ZSUBF

Table 2.3-17 TesterPresent Request Message

2.3.2.3.3 Positive Response Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	testerPresent Positive Response Service Id	S	7E	TPPR
#2	Response Required = [zeroSubFunction]	M	00	ZSUBF

Table 2.3-18 TesterPresent Positive Response Message

2.3.2.3.4 Negative Response Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	testerPresent Request Service Id	M	3E	TP
#3	ResponseCode = [subFunctionNotSupported incorrectMessageLengthOrInvalidFormat]	M	 12 13	RC SFNS IMLOIF

Table 2.3-19 TesterPresent Negative Response Message

Hex	Conditions
12	subFunctionNotSupported 1. Requested sub-function is not supported
13	incorrectMessageLengthOrInvalidFormat 1. Wrong length for the specific request

Table 2.3-20 Diagnostic Services $3E Response Code

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

2.3.3 Data Transmission functional unit

Service name	Description
ReadDataByIdentifier	The client requests to read the current value of a record identified by a provided dataIdentifier.
ReadMemoryByAddress	The client requests to read the current value of the provided memory range.
WriteDataByIdentifier	The client requests to write a record specified by a provided dataIdentifier.
Test	This is an OEM specified Service. This client sends data in its request message and the server simply take the data record from the request and return it in the response message.

Figure 2.3-2 Transmission functional unit

2.3.3.1 ReadDataByIdentifier service (SId $22)

The ReadDataByIdentifier service allows the client to request data record values from the server identified by one or more dataIdentifiers.

The server limits the number of dataIdentifiers that can be simultaneously requested as 1.

Upon receiving a ReadDataByIdentifier request, the server will access the data elements of the records specified by the dataIdentifier parameter(s) and transmit their value in one single ReadDataByIdentifier positive response containing the associated dataRecord parameter(s).

2.3.3.1.1 Request Message definition

2.3.3.1.1.1 Sub-function Parameter Level definition

This service does not use a sub-function parameter.

2.3.3.1.1.2 Data parameter definition

For supported data-parameters please see the section 4.3 Request Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	ReadDataByIdentifier Request Service Id	M	22	RDBLI
#2 #3	dataIdentifier = [byte#1 (MSB) byte#2]	M	 00-FF 00-FF	DID HB LB

Table 2.3-21 ReadDataByIdentifier Request Message

2.3.3.1.2 Positive Response Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	ReadDataByIdentifier Request Service Id	S	62	REIPR
#2 #3	dataIdentifier =[byte#1 (MSB) byte#2]	M	 00-FF 00-FF	DID HB LB

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

#4 : #n	datarecord#1 = [data#1 : data#k]	M : C	xx, : xx	DREC_ DATA_1 : DATA_k

Table 2.3-22 ReadEcuIdentification Positive Response Message

2.3.3.1.3 Negative Response Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	ReadDataByIdentifier Request Service Id	M	21	REI
#3	ResponseCode = [incorrectMessageLengthOrInvalidFormat requestOutOfRange]	M	 13 31	 IMLOIF ROOR

Table 2.3-23 ReadDataByIdentifier Negative Response Message

Hex	Conditions
13	incorrectMessageLengthOrInvalidFormat 1. Wrong length for the specific request
31	RequestOutOfRange This response code will be sent if: 1. Any DataIdentifier in the request is invalid (reserved or not defined).

Table 2.3-24 Diagnostic Services $21 Response Code

2.3.3.2 ReadMemoryByAddress service (SId $23)

The ReadMemoryByAddress service allows the client to request memory data from the server via provided starting address and size of memory to be read.
The readMemoryByAddress request message requests memory data from the server identified by the parameters memoryAddress and memorySize.

This diagnostic service will be used to read most of RAM, some portion of flash memory in a physically address request. DBC 7.8 will deny access to any other memory address.

2.3.3.2.1 Request Message definition

2.3.3.2.1.1 sub-function Parameter Level definition

This service does not use a sub-function parameter.

2.3.3.2.1.2 data parameter definition

The following data-parameters are defined for this service.

Definition
addressAndLengthFormatIdentifier This parameter is a one byte value with each nibble encoded separately bit 7 - 4: number of bytes of the memorySize parameter. Support 1 byte format bit 3 - 0: number of bytes of the memoryAddress parameter. Support 3 byte format
memoryAddress The parameter memoryAddress is the starting address of server memory from which data is to be retrieved. The number of bytes used for this address is defined by the low nibble (bit 3 - 0) of the addressFormatIdentifier. Byte#3 in the memoryAddress parameter is always the least significant byte of the address being referenced in the server.
memorySize

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

The parameter memorySize in the ReadMemoryByAddress request message specifies the number of bytes to be read starting at the address specified by memoryAddress in the server's memory. The number of bytes used for this size is defined by the high nibble (bit 7 - 4) of the addressFormatIdentifier. The maximum number of bytes can be requested is 31 bytes.

2.3.3.2.2 Request Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	readMemoryByAddress Request	M	23	RMBA
#2	addressAndLengthFormatIdentifier	M	13	ALFID
#3 #4 #5	memoryAddress[] = [byte#1 (MSB) byte#2 byte#3]	M : C_1	00-FF 00-FF 00-FF	MA_ B1 B2 B3
#6	memorySize =[byte#1]	M	01-1F	MS B1

Table 2.3-25 ReadMemoryByAddress Request Message

2.3.3.2.3 Positive Response Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	readMemoryByAddress Positive Response	S	63	RMBAPR
#2 : #n	dataRecord[] = [data#1 : data#m]	M : U	00-FF : 00-FF	DREC_ DATA_1 : DATA_m

Table 2.3-26 ReadMemoryByAddress Positive Response Message

2.3.3.2.4 Negative Response Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	readMemoryByAddressRequest Service Id	M	23	RMBA
#3	ResponseCode = [incorrectMessageLengthOrInvalidFormat requestOutOfRange]	M	13 31	IMLOIF ROOR

Table 2.3-27 ReadMemoryByAddress Negative Response Message

Hex	Conditions
13	<u>SubFunctionNotSupported-InvalidFormat</u> 1. Request message has wrong bytes count.
31	<u>RequestOutOfRange</u> 1. Any memory address within the interval [$MA, ($MA + $MS -$1)] is invalid (i.e. not in RAM or readable FlashMemory) or restricted. 2. The memorySize parameter in the request message is greater than the maximum value supported by the ECU. 3. The specified addressAndLengthFormatIdentifier is not valid

Table 2.3-28 Diagnostic Services $23 Response Code

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

2.3.3.3 WriteDataByIdentifier service (SId $2E)

The WriteDataByIdentifier service allows the client to write information into the server at an internal location specified by the provided data identifier. The WriteDataByIdentifier service is used by the client to write a dataRecord to a server. The data is identified by a dataIdentifier, and may or may not be secured. It is the vehicle manufacturer's responsibility that the server conditions are met when performing this service.

Possible uses for this service : This service is only supported if requested by Physical address mode.

2.3.3.3.1 Request Message definition

2.3.3.3.1.1 sub-function Parameter Level definition

This service does not use a sub-function parameter.

2.3.3.3.1.2 data parameter definition

The following data-parameters are defined for this service.

dataIdentifier This parameter identifies the server data record that the client is requesting to write to
dataRecord This parameter provides the data record associated with the dataIdentifier that the client is requesting to write to.

Table 2.3-29 — Request message data parameter definition

For detailed dataParameter please see 4.3.3 DataIdentifier Definition P. 55

2.3.3.3.2 Request Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	WriteDataByIdentifier Request Service Id	M	2E	WDBLI
#2 #3	dataIdentifier [] =[byte#1(MSB) byte#2]	M	 00-FF 00-FF	DID_ HB LB
#4 : #n	dataRecord[] = [data#1 : data#m]	 M : M	 xx : xx	DREC_ DATA_1 : DATA_m

Table 2.3-30 WriteDataByIdentifier Request Message

2.3.3.3.3 Positive Response Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	WriteDataByIdentifier Positive Response Service Id	S	6E	WDBLIPR
#2 #3	dataIdentifier [] =[byte#1(MSB) byte#2]	M	 00-FF 00-FF	DID_ HB LB

Table 2.3-31 WriteDataByIdentifier Positive Response Message

2.3.3.3.4 Negative Response Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	WriteDataByIdentifier Request Service Id	M	2E	WDBLI

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

#3	ResponseCode = [incorrectMessageLengthOrInvalidFormat ConditionsNotCorrect requestOutOfRange securityAccessDenied generalProgrammingFailure reqCorrectlyRcvd-RspPending serviceNotSupportedInActiveSession]	M	 13 22 31 33 72 78 7F	 IMLOIF CNC ROOR SAD GPF RCR-RP SNSIAS

Table 2.3-32 WriteDataByIdentifier Negative Response Message

Hex	Conditions
13	incorrectMessageLengthOrInvalidFormat 1. This response code will be sent if the length of the data in the request message does not match the size expected by the node.
22	ConditionsNotCorrect 1. This response code shall be sent if the operating conditions of the ECU are such that it can't perform the required action (e.g., the data for a DataIdentifier is stored in EEPROM and an EEPROM failure has occurred). 2. This response code will occur if the node (ECU) internal conditions do not allow the writing of any DataIdentifier/ IdentificationOption. • The system is currently in initialization. • The system is currently in shutdown process. • The system is currently in SafetySystemDiagnosticSession. • The system is currently in InputOutputControlByIdentifier service active stage. • The system is currently busy at modifying memory. • The vehicle is moving.
31	RequestOutOfRange This response code will be sent if: 1. The DataIdentifier in the request message is not supported.
33	securityAccessDenied This negative response code is used for when security is not yet granted to perform the required service.
72	generalProgrammingFailure 1. Request can not be serviced or system has error while try to change the EEPROM contents.
78	ReqCorrectlyRcvd-RspPending 1. This response code shall be sent if writing the data to the address(es), which are referenced by the DataIdentifier, takes longer than $P2_{can_server}$ ms.
7F	serviceNotSupportedInActiveSession 1. The service is not enabled in the current diagnostic session.

Table 2.3-33 Diagnostic Services $2E Response Code

2.3.3.4 Test service (SId $B0)

This service is defined by OEM. It is only used for specific test purposes.
This service simply take the data record from the request and return it in the response message. The length of the requestRecord may be chosen to suit the test to be performed. Maximum length of the message should not exceed 30 bytes in order to not impose additional size of message buffer area.

2.3.3.4.1 Request Message definition

2.3.3.4.1.1 Sub-function Parameter Level definition

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

This service has one sub-function parameter.

Hex (bit 6-0)	Description	Mnemonic
01	**Echo request** The server takes recorded data from the client request and returns it in the response message.	**ER**

Table 2.3-34 Request message sub function definition

2.3.3.4.1.2 Data parameter definition

This service does not support data parameters in the request message.

2.3.3.4.2 Request Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	TestService request service Service Id	M	B0	TS
#2	sub-function= [Echo request]	M	01	ER
#3 : #n	RequestRecord = [data#1 : data#m]	U : U	00~FF : 00~FF	RR DATA_1 : DATA_m

Table 2.3-35 TestService Request Message

2.3.3.4.3 Positive Response Message

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	TestService positive response service Id, SIDPR	**S**	**F0**	**TSPR**
#2	sub-function= [Echo request]	M	01	**ER**
#3 : #n	RequestRecord = [data#1 : data#m]	C : C	00~FF : 00~FF	RR DATA_1 : DATA_m

Table 2.3-36 TestService Positive Response Message

C: depend on RequestRecord in request message. It should have the same content and length as the request message when it is short or equal to 30 bytes of data. (m <= 30)

2.3.3.4.4 Negative Response Message

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	TestService Request Service Id	M	B0	WDBLI
#3	ResponseCode = [subFunctionNotSupported incorrectMessageLengthOrInvalidFormat ConditionsNotCorrect serviceNotSupportedInActiveSession]	M	12 13 22 7F	SFNS IMLOIF CNC SNSIAS

Table 2.3-37 Testservice Negative Response Message

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Hex	Conditions
12	subFunctionNotSupported Requested sub-function is not supported
13	incorrectMessageLengthOrInvalidFormat Request has Data longer than 30 bytes
22	ConditionsNotCorrect Request is functional addressed
7F	serviceNotSupportedInActiveSession The service is not enabled in the current diagnostic session.

Table 2.3-38 Diagnostic Services $B0 Response Code

2.3.4 Stored Data Transmission functional unit

Service name	Description
ClearDiagnosticInformation	The client requests from the server to clear all or a group of the diagnostic information stored.
ReadDiagnosticTroubleCodesBy-Status	The client requests from the server the transmission of both, number of DTC and values of the diagnostic trouble codes depending on their status.

Table 2.3-39 Stored Data Transmission functional unit

2.3.4.1 ClearDiagnosticInformation service (SId $14)

The clearDiagnosticInformation service is used by the client (tester) to clear diagnostic information in the server's (ECU's) memory. If the parameter the server shall send a positive response when the ClearDiagnosticInformation service is completely processed. The server shall send a positive response even if no DTC's are stored. If a server supports multiple copies of DTC status information in memory (e.g. one copy in RAM and one copy in EEPROM), the server shall clear the copy used by the ReadDTCInformation status reporting service followed by the remaining copy.

This service will clear the DTC status bit(s) and failure Record Information stored in RAM and permanent memories immediately after receiving the valid ClearDiagnosticInformation request. A system reset will be triggered by the valid request. The parameters used for Fault Detection; e.g. timers, flags and variables are also reset. The amount of time taken to clear the diagnostic related information depends on the stored fault information. If the conditions responsible for fault detection are still present after this request is processed, the fault will be logged again.

2.3.4.1.1 Request Message definition

2.3.4.1.1.1 Sub-function Parameter Level definition

This service does not use a sub-function parameter.

2.3.4.1.1.2 Data parameter definition

The following data-parameters are defined for this service.

groupOfDTC This parameter contains a 3-byte value indicating the group of DTC's (e.g., Powertrain, Body, Chassis) or the particular DTC to be cleared.

Table 2.3-40 — Request message data parameter definition

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

2.3.4.1.2 Request Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	clearDiagnosticInformation Request Service Id	M	14	CDTCI
#2 #3 #4	groupOfDTC[] = [groupOfDTCHighByte groupOfDTCMiddleByte groupOfDTCLowByte]	M	 00-FF* 00-FF* 00-FF*	GODTC_ HB MB LB
* All group FF FF FF, chassis C1F FF FF group, any DTC in C1 group, will be possible to get a positive response. Any other, will get negative response.				

Table 2.3-41 ClearDiagnosticInformation Request Message

2.3.4.1.3 Positive Response Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	clearDiagnosticInformation Positive Response Sid	S	54	CDIPR

Table 2.3-42 ClearDiagnosticInformation Positive Response Message

2.3.4.1.4 Negative Response Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	clearDiagnosticInformation Request Service Id	M	14	CDTCI
#3	ResponseCode = [generalReject incorrectMessageLengthOrInvalidFormat ConditionsNotCorrect requestOutOfRange reqCorrectlyRcvd-RspPending reqCorrectlyRcvd-RspPending]	M	 10 13 22 31 78 7F	 GR IMLOIF CNC ROOR RCR-RP SNSIAS

Table 2.3-43 ClearDiagnosticInformation Negative Response Message

Hex	Conditions
10	generalReject Request can not be serviced system has error while try to clear fault information.
13	incorrectMessageLengthOrInvalidFormat Request has wrong length.
22	ConditionsNotCorrect This response code will occur if the node (ECU) internal conditions do not allow deletion of the diagnostic information stored in the node. These are: • The vehicle is moving (about 5 kph); • The system is currently in initialization or shutdown process. • The system is currently in SafetySystemDiagnosticSession. • The system is currently in I/O Control. • The system is in the process of changing some Memory content.
31	requestOutOfRange Wrong groupOfDTC values in request
78	reqCorrectlyRcvd-RspPending

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Hex	Conditions
	System is busy processing the clear code request which takes longer than $P2_{can_server}$ ms.
7F	serviceNotSupportedInActiveSession The service is not enabled in the current diagnostic session.

Table 2.3-44 Diagnostic Services $14 Response Code

2.3.4.2 ReadDTCInformation service (SId $19)

This service allows a client to read the status of server resident Diagnostic Trouble Code (DTC) information from any server, or group of servers within a vehicle. Unless otherwise stated, the server shall return DTC information. In this service we allow the client to do the following:

- Retrieve the number of DTC's matching a client defined DTC status mask (at the point of the request),
- Retrieve the list of all DTC's matching a client defined DTC status mask,
- Retrieve the status of all DTC's supported by the server,

This service uses a sub-function to determine which type of diagnostic information the client is requesting. Further details regarding each sub-function parameter are provided in the following sections.

Note:

- Diagnostic trouble codes being reported under this service are not ordered by the time they were detected.
- When there are no DTC's stored the ECU will also return a single positive response.

2.3.4.2.1 Request Message definition

2.3.4.2.1.1 Sub-function Parameter Level definition

The sub-function parameters are used by this service to select one of the DTC report types specified in Table 2.3-45 Request message sub function definition. Explanations and usage of the possible levels are detailed below (suppressPosRspMsgIndicationBit (bit 7) not shown).

Hex (bit 6-0)	Description	Mnemonic
01	**reportNumberOfDTCByStatusMask** This parameter specifies that the server shall transmit to the client the number of DTC's matching a client defined status mask.	RNODTCBSM
02	**reportDTCByStatusMask** This parameter specifies that the server shall transmit to the client a list of DTC's and corresponding statuses matching a client defined status mask.	RDTCBSM

Table 2.3-45 Request message sub function definition

2.3.4.2.1.2 Data parameter definition

The following data-parameters are defined for this service.

Definition
DTCStatusMask The DTCStatusMask contains eight (8) DTC status bits. The definitions for each of the eight (8) bits can be found in Table 2.3-47 Definition of statusOfDTC values. This byte is used in the request message to allow a client to request DTC information for the DTC's whose status matches the DTCStatusMask. A DTC's status matches the DTCStatusMask if any one of the DTC's actual status bits is set to '1' and the corresponding status bit in the DTCStatusMask is also set to '1' (i.e., if the DTCStatusMask is bit-wise logically ANDed with the DTC's actual status and the result is non-zero, then a match has occurred). If the client specifies a status mask that contains bits that the server does not support, then the server shall process the DTC information using only the bits that it does support.

Table 2.3-46 — Request data parameter definition

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

2.3.4.2.2 DTC status bit definitions

Bit	Description of statusOfDTC-Response	Mnemonic	Bit support
0	testFailed	WIRS	Yes
1	testFailedThisOperationCycle	CDTCSPU	Yes
2	pendingDTC	TNPSCPU	No
3	confirmedDTC	HDTC	Yes
4	testNotCompletedSinceLastClear	TFSDTCC	Yes
5	testFailedSinceLastClear	TNPSDTCC	Yes
6	testNotCompletedThisOperationCycle	CDTC	Yes
7	warningIndicationRequested	DTCSBC	No

Table 2.3-47 Definition of statusOfDTC values

For more detail of the DTC status bit definition, please check ISO 14229 annex D.

2.3.4.2.3 Request Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	ReadDTCInformation request Service Id	M	19	RDTCI
#2	sub-function = [reportNumberOfDTCByStatusMask reportDTCByStatusMask]	M	 01 02	LEV_ RNODTCBSM RDTCBSM
#3	DTCStatusMask	M	00-FF	DTCSM

Table 2.3-48 ReadDiagnosticTroubleCodesByStatus Request Message

2.3.4.2.4 Positive Response Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	ReadDTCInformation response Service Id	M	59	RDTCIPR
#2	reportType = [reportNumberOfDTCByStatusMask]	M	01	LEV_ RNODTCBSM
#3	DTCStatusAvailabilityMask	M	7B	DTCSAM
#4	DTCFormatIdentifier = [ISO14229-1DTCFormat]	M	01	DTCFID_ 14229-1DTCF
#5 #6	DTCCount[] = [DTCCountHighByte DTCCountLowByte]	M M	00-FF 00-FF	DTCC_ DTCCHB DTCCLB

Table 2.3-49 ReadDiagnosticTroubleCodesByStatus Positive Response Message

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	ReadDTCInformation response Service Id	M	59	RDTCIPR
#2	reportType = [reportDTCByStatusMask]	M	02	LEV_ RDTCBSM
#3	DTCStatusAvailabilityMask	M	7B	DTCSAM

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
	DTCAndStatusRecord[] = [			DTCASR_
#4	DTCHighByte#1	C_1	00-FF	DTCHB
#5	DTCMiddleByte#1	C_1	00-FF	DTCMB
#6	DTCLowByte#1	C_1	00-FF	DTCLB
#7	statusOfDTC#1	C_1	00-FF	SODTC
#8	DTCHighByte#2	C_2	00-FF	DTCHB
#9	DTCMiddleByte#2	C_2	00-FF	DTCMB
#10	DTCLowByte#2	C_2	00-FF	DTCLB
#11	statusOfDTC#2	C_2	00-FF	SODTC
:	:	:	:	:
#n-3	DTCHighByte#m	C_2	00-FF	DTCHB
#n-2	DTCMiddleByte#m	C_2	00-FF	DTCMB
#n-1	DTCLowByte#m	C_2	00-FF	DTCLB
#n	statusOfDTC#m]	C_2	00-FF	SODTC
C_1:	This parameter is only present if DTC information is available to be reported.			
C_2:	This parameter is only present if more than one DTC information is available to be reported.			

Table 2.3-50 — Response message definition - sub-function = reportDTCByStatusMask, reportSupportedDTCs

2.3.4.2.5 Negative Response Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	readDiagnosticTroubleCodesByStatus Request SId	M	19	RDTCI
#3	ResponseCode = [subFunctionNotSupported incorrectMessageLengthOrInvalidFormat reqCorrectlyRcvd-RspPending]	M	 12 13 78	 SFNS IMLOIF RCR-RP

Table 2.3-51 ReadDiagnosticTroubleCodesByStatus Negative Response Message

Hex	Conditions
12	subFunctionNotSupported Requested sub-function is not supported
13	incorrectMessageLengthOrInvalidFormat Wrong length for the specific request
78	ReqCorrectlyRcvd-RspPending This response code shall be sent if writing the data to the address(es), which are referenced by the DataIdentifier, takes longer than $P2_{can_server}$ ms.

Table 2.3-52 Diagnostic Services $18 Response Code

2.3.5 InputOutput Control functional unit

The services provided by this functional unit are described in table 3.5:

Service name	Description
InputOutputControlByLocalIdentifier	The client requests the control of an input/output specific to the server.

Table 2.3-53 InputOutput Control functional unit

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

2.3.5.1 InputOutputControlByIdentifier (Sid $2F)

The InputOutputControlByIdentifier service is primarily used by DBC 7.8 to control some of the outputs (actuators) of the electronic system. In another word, it is used as Device Control.
The client request message contains a dataIdentifier to reference a group of signals of the server.
The server shall send a positive response message if the request message can be successfully executed.

Exiting Device Control

Some of the components excited by I/O Control commands will remain active for a while (settling time) even after they have been de-activated by I/O Control command. When any of the following conditions are met, the system will begin to terminate this service. The system will, however, remain in the I/O Control until it is determined that the commanded component(s) is / are no longer active. The system will then return control to normal operation. During the component settling time, the system is allowed to go back to this service. The exiting conditions are:

1. Positively responded to a new diagnostic session request.
2. Any Output Control limit is exceeded. See section 2.3.5.1.4 Negative Response Message Data Bytes
3. The active Session is timed out per S3server.

This service is only supported if requested by Physical address mode.

2.3.5.1.1 Request Message definition

2.3.5.1.1.1 Sub-function Parameter Level definition

This service does not use a sub-function parameter.

2.3.5.1.1.2 Data parameter definition

The following data-parameters are defined for this service.

Definition
dataIdentifier This parameter identifies a server local input signal(s), internal parameter(s) or output signal(s). The applicable range of values for this parameter can be found in the table of dataIdentifiers defined in Section 4.3.3 DID $3100 ~ $311F DeviceControl. In our implementation the high byte of the dataIdentifier is used as the “Control Packet ID” or CPID and the low byte is used as the control enable bytes. However it is not used as a enable control of the particular device through this service but a control ON enable byte.
controlOptionRecord The controlOptionRecord of each dataIdentifier consists of one or multiple bytes (controlState#1 to controlState #m). ControlState#1 can be used as an additional controlState byte

Table 2.3-54 Request message data parameter definition

For detailed information on the dataIdentifer byte #2 and ControlOptionRecord#1 please see section 4.3.3

2.3.5.1.2 Request Message Data Bytes

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	**inputInputOutputControlByIdentifier Request Sid**	**M**	**2F**	**IOCBLI**
	dataIdentifier #1 [] = [	M		**IOI**
#2	byte #1 (MSB)		31	**B1**
#3	byte #2 (LSB)]		00-FF	**B2**
#4	ControlOptionRecord #1 = [			**CSR**
#5	ControlStatus#1	M	00-FF	**CS**
	ControlStatus#2	M	00-FF	**CS**
	ControlStatus#3	M	00-FF	**CS**
	ControlStatus#4]	M	00-FF	**CS**

Table 2.3-55 InputOutputControlByIdentifier request message

2.3.5.1.3 Positive Response Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	inputInputOutputControlByIdentifier Positive Response Sid	S	6F	IOCBLIPR
#2	dataIdentifier #1 [] = [	M		IOI
#3	byte #1 (MSB)		31	B1
	byte #2 (LSB)		00-FF	B2
	]			

Table 2.3-56 InputOutputControlByIdentifier positive response message

2.3.5.1.4 Negative Response Message Data Bytes

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	inputInputOutputControlByIdentifier Request SId	M	2F	IOCBLI
#3	ResponseCode = [	M		
	incorrectMessageLengthOrInvalidFormat		13	IMLOIF
	ConditionsNotCorrect		22	CNC
	RequestOutOfRange		31	ROOR
	serviceNotSupportedInActiveSession		7F	SNSIAS
	VehicleSpeedLimitExceeded		88	VSLE
	MotorOrSolenoidThermalLimitExceeded		86	MOSTLE
	SystemVoltageOutOfRange		93	SVOOR
	]			

Table 2.3-57 InputOutputControlByIdentifier negative response message

Hex	Condition
13	<u>incorrectMessageLengthOrInvalidFormat</u> Wrong length for the specific request
22	<u>ConditionsNotCorrect</u> This response code shall be sent if the operating conditions of the ECU are such that it can't perform the required action. • The system is currently in initialization or shutdown process. • The system is in the process of changing memory. .
31	<u>RequestOutOfRange</u> Wrong inputOutputLocalIdentifier
7F	<u>serviceNotSupportedInActiveSession</u> The service is not enabled in the current diagnostic session.

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

88	VehicleSpeedLimitExceeded 1. This code will be returned any time the device exits I/O Control due to an unsafe condition. That is, the vehicle speed ≥ 5 KPH, while the device's is not currently in SafetySystemDiagnosticSession. Please note that this response can be send unsolicited.
86	MotorOrSolenoidThermalLimitExceeded 1. This code will be returned any time the device exits I/O Control due to its internal protection algorithms. That is, any of the solenoids or motor has been commanded active for an excessive period of time that may begin to damage the device. 2. Please note that this response can be send unsolicited.
93	SystemVoltageOutOfRange 1. This code will be returned any time the device exits I/O Control due to the ignition voltage not been sufficient 2. Please note that this response can be send unsolicited.

Table 2.2.4.1.5-7 Diagnostic Services $2F Response Code

3 Service Scan Tool Interface

The following section describes recommended functions that should be implemented in service scan tools that will be used with the DBC 7.8 PM EHCU. All of these functions utilize the ISO14229 diagnostic services that are described Section 2.0

3.1 Service Scan Tool Communications with Controller

In order to start and maintain diagnostic communication with the service scan tool and the DBC 7.8 PM EHCU must receive service request at a period less than 5 seconds.

3.2 Diagnostic Trouble Code Information

The service scan tool should support the following functions for obtaining diagnostic code information from the DBC 7.8 PM EHCU.

3.2.1 Read Current Diagnostics Codes

The service scan tool should have the capability to read diangostic codes using ISO14229 ReadDiagnosticTroubleCodesByStatus service - Sid $19. (See Section 2.3.4.2.) Note that the response diagnostic trouble code information will utilize the trouble code numbers that are include the DBC78 Customer Diagnostic Code Cross Reference Table.

3.2.2 Read Enhanced Diagnostic History

The DBC 7.8 PM EHCU provides enhanced diagnostic history functions that will record various vehicle and brake system parameters at the time a fault code is set. The service scan tool should have the capability to read the enhanced diagnostic codes information using the readDataByIdentifier service Sid $22 (see Section 2.3.3.1) and the corresponding supplier defined DIDSs for the Last, Second, Third, Fourth, and Fifth fault occurrences.. See Section 4.3.2 for additional description of the parameters provided by this service.

3.2.3 Clear Diagnostic Codes

The service scan tool should have the capability to clear diagnostic codes using the ISO14229 ClearDiagnosticInformation service - Sid $14. (See Section 2.3.4.1)

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

3.3 ECU Identification

The service scan tool should have the ability to obtain the part number information that is stored within the DBC 7.8 PM EHCU memory. This part number information is used to identify the specific DBC78 hardware and software that are installed on the vehicle. This information can be obtained using the static data identifiers described in section an the readDataByIdentifier servce Sid $22 (see section 2.3.3.1)

3.4 Data Display Functions

The service scan tool should support data display functions that can show near-realtime data from the DBC78 EHCU. Such data is useful for root causing DTCs from the DBC78 EHCU and other related vehicle systems. The scan tool should also have the capability to record, store, and playback data from the DBC78 EHCU. Data recording functions (also called 'Snap Shot" functions) enable the scan tool user to record and view general stauts data, and input / ouput signals when investigating customer complaints or diagnostic trouble codes Such functions can also be used to confirm the effectivensss of a repair after work has been completed.

The data display functions can be implemented in a scan tool using the ReadDataByIdentifier service - SId $22 and the supplier defined dynamic identifiers described in section 4.3.2. This data is available under all vehicle normal vehicle operating conditions.

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

3.5 Input / Output Device Control

Input / Output Device Control (referred to as “Device Control”) enables the service technician to manually control the outputs of the DBC 7.8 PM EHCU. Within the ISO14229 diagnostic service protocol, these functions are performed using InputOutputControlByLocalIdentifier service SId $2F and the corresponding data indentifiers described in section 4.3.3. The device control functions are used for normal vehicle assembly and service operations on the vehicle’s brake system. Device control can also be used to investigate and root cause vehicle service issues that are caused by either the DBC78 EHCU or other vehicle systems.

Implementation of the device control features within a service scan tool can greatly improve the quality and ease of service operations. The InputOutputControlByLocalIdentifier service for the DBC78 PM EHCU is implemented using the following message format.

It is important to note that when a technician uses device control functions on the DBC78 EHCU, normal ABS functions will not be available since some device control commands will restrict brake fluid flow to the the wheel brakes. Refer DID Section 4.3.3 for control status bytes details.

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

3.5.1 Hydraulic Output Control

3.5.1.1 Solenoid Control

Device control can be used to manually energize the solenoid valves in the DBC78 PM EHCU. To energize any of the DBC78 valves, the Sid $2F message must include the following data:
Bit 0 and 2 in DID byte #2 should be enabled and bit 3 is optional.
For any of the solenoids, individual bits in control state byte should be set to 1/0 for Solenoid ON/OFF control.

3.5.1.2 Pump Control

Device control can be used to manually turn-on the EBCM pump. To actuate any of the DBC78 pump, the Sid$2F message must include the following data:

Bit 0 and 1 in DID byte #2 should be enabled.
Bit 1 Control state byte #2 can be set to 1/0 for Motor ON/Off control.

3.5.1.3 Brake System Bleed

The solenoid and pump control features can be utilized to create a brake system bleed routine. Refer to the DBC78 Diagnostic and Service Manual for a detailed description of the bleed procedure.

The table below show a series of Sid$2F message requests that cycle the left front and right rear valves while running the pump (Important: This is an example of the Sid $2F commands – it is NOT an acceptable brake bleed procedure).

Control Action	Sid$2F Message Data Bytes (Hex)									
	1	2	3	4	5	6	7	8	9	10
Energize LF and RR Release valves and run the pump	2F	31	7	82	3	0	0	0	0	0
De-Energize LF and RR Release valves, continue running pump	2F	31	7	0	3	0	0	0	0	0
Energize LF Apply valve and run the pump	2F	31	7	1	3	0	0	0	0	0
De-Energize LF Apply valve while running the Pump	2F	31	7	0	3	0	0	0	0	0
Energize RR Apply valve while running the Pump	2F	31	7	40	3	0	0	0	0	0
De-energize RR Apply valve while running the Pump	2F	31	7	0	3	0	0	0	0	0
End Control	2F	31	0	0	0	0	0	0	0	0

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

4 Parameter Definitions

4.1 Parameter Coding

The parameters shall be coded with data types defined ASCII character set according to SAE2178-2.

4.2 Bit and Byte Order

Bit order: According to ISO/DIS 11898:1993/Amd.1

Byte order: Multiple byte parameters shall always be sent with the most significant byte first. For example, If a variable is two-byte long, at Data Byte #k, place the most significant byte, at Data Byte #k+1, place the least significant byte.

4.3 DataIdentifier Definition

The parameter Data identifier is used by the ReadDataByIdentifier service (SId $22) to request static or dynamic data from the server, by WriteDataByIdentifier service (SId $2E) to write data into the server or by InputOutputControlByIdentifier (Sid $2F) to device control the outputs. The $FXXX range specified by ISO is used by OEM to define the vehicle common dataIdentifier. The $30XX is OEM designated range for ABS to define the static or dynamic dataIdentifier. The $31XX is OEM designated range for ABS to use for device control service. DIDs for a specific program are a subset from the following table. Read and write access to a DID is also program dependent and customer defined. The list is not all inclusive of DIDs that are supported in the product.
If a variable is two-byte long, then place the MSB at the lower Data Byte count. For example, at Data Byte #k, place variable_a MSB, at Data Byte #k+1, place variable_a LSB.

Optional dataIdentifier	Definition	Read	Write
0100	OEMSpecific_1	X	
F182	applicationDataIdentificationDataIdentifier	X	X
F183	ECUBootloaderSoftwareReferenceNumber	X	
F187	ECUPartNumber	X	X
F18A	SystemSupplierIdentifier	X	
F18B	ECUManufactureDate	X	
F190	vehicleIdentificationNumber	X	X
F192	systemSupplierECUHardwareNumber	X	
F193	systemSupplierECUHardwareVersionNumber	X	
F194	systemSupplierECUSoftwareNumber	X	
F195	SystemSupplierECUSoftwareVersionNumberDataIdentifier	X	
F197	SystemName	X	
F199	DateECUReprogrammed	X	X
F19B	CalibrationDateInformation	X	X
F19D	ECUInstallationDateDataIdentifier	X	X
F1A0	OEMSpecific_2	X	X
F1A1	OEMSpecific_3	X	X
3003	SupplierEHCUTraceability	X	X
3007	System Speed and Status Information	X	
3008	Power and Solenoid Information	X	
3009	Current And History ECU MCU DTC Information	X	
300A	Last Fault Diagnostic Information	X	

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

300B	Second Last Fault Diagnostic Information	X	
300C	Third Last Fault Diagnostic Information	X	
3017	4th Last Fault Diagnostic Information	X	
3018	5th Last Fault Diagnostic Information	X	
300D	CurrentDiagnosticDTCInformation	X	
300E	HistoryDiagnosticDTCInformation	X	
3020	ManufacturersEnableCounter	X	X
3100 to 311F	DeviceControl	n/a	n/a

Table 4.3-1 DataIdentification Parameter definition

X – indicates Read or Read/Write capable.

4.3.1 Static DataIdentifier

The data values and format for above table items is exemplified in the following sections.

4.3.1.1 DID $0100 OEMSpecific_1 (Read/Write)

DID Byte Index	Definition	Display
#1	OEM Specific_1 – 1st char	ASCII
#2	OEM Specific_1 – 2nd char	ASCII
#3	OEM Specific_1 – 3rd char	ASCII
#4	OEM Specific_1 – 4th char	ASCII
#5	OEM Specific_1 – 5th char	ASCII
#6	OEM Specific_1 - 6th char	ASCII
#7	OEM Specific_1 – 7th char	ASCII
#8	OEM Specific_1 – 8th char	ASCII
#9	OEM Specific_1 – 9th char	ASCII
#10	OEM Specific_1 - 10th char	ASCII
#11	OEM Specific_1 - 11th char	ASCII
#12	OEM Specific_1 - 12th char	ASCII
#13	OEM Specific_1 – 13th char	ASCII
#14	OEM Specific_1 – 14th char	ASCII
#15	OEM Specific_1 – 15th char	ASCII
#16	OEM Specific_1 – 16th char	ASCII

4.3.1.2 DID $F182 applicationDataIdentificationDataIdentifier (Read/Write)

DID Byte Index	Definition	Display
#1	Application Data Identification – 1st char	ASCII
#2	Application Data Identification – 2nd char	ASCII
#3	Application Data Identification – 3rd char	ASCII
#4	Application Data Identification – 4th char	ASCII
#5	Application Data Identification – 5th char	ASCII
#6	Application Data Identification - 6th char	ASCII
#7	Application Data Identification – 7th char	ASCII
#8	Application Data Identification – 8th char	ASCII
#9	Application Data Identification – 9th char	ASCII
#10	Application Data Identification - 10th char	ASCII
#11	Application Data Identification - 11th char	ASCII
#12	Application Data Identification - 12th char	ASCII
#13	Application Data Identification – 13th char	ASCII
#14	Application Data Identification – 14th char	ASCII

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

DID Byte Index	Definition	Display
#15	Application Data Identification – 15th char	ASCII
#16	Application Data Identification – 16th char	ASCII

4.3.1.3 DID $F183 ECUBootloaderSoftwareReferenceNumber (Read)

DID Byte Index	Definition	Display
#1	ECU Boot loader Part Number MSB	ASCII
#2	ECU Boot loader Part Number 2nd MSB	ASCII
#3	ECU Boot loader Part Number	ASCII
#4	ECU Boot loader Part Number	ASCII
#5	ECU Boot loader Part Number	ASCII
#6	ECU Boot loader Part Number	ASCII
#7	ECU Boot loader Part Number	ASCII
#8	ECU Boot loader Part Number	ASCII
#9	ECU Boot loader Part Number 2nd LSB	ASCII
#10	ECU Boot loader Part Number LSB	ASCII

- Note: If spare bytes, please set 0 in LSB bytes

4.3.1.4 DID $F187 ECUPartNumber (Read/Write)

This version of the DID is utilized by applications with a part number configuration that requires up to n byte = 16 bytes. Length is variable based upon the part number length. The interpretation of the byte values (** HEX, USN, or ASCII) will depend on the customer part number format.

DID Byte Index	Definition	Display
#1	35	BCD
#2	50	BCD
#3	01	BCD
#4	00	BCD
#5	01	BCD
#6	2D	ASCII
#7	45	ASCII
#8	30	ASCII
#9	31	ASCII

**** Value will be HEX, USN, or ASCII as required by application.**

4.3.1.5 DID $F18A systemSupplierIdentifier (Read)

DID Byte Index	Definition	Display
#1	32	ASCII
#2	37	ASCII
#3	31	ASCII
#4	31	ASCII
#5	54	ASCII

4.3.1.6 DID $F18B ECUManufactureDate(Read)

DID Byte Index	Definition	Display
#1	Year	BCD

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

DID Byte Index	Definition	Display
#2	Month	BCD
#3	Date	BCD

4.3.1.7 DID $F190 vehicleIdentificationNumber (Read/Write)

DID Byte Index	Definition	Display
#1	VIN – 1st char	ASCII
#2	VIN – 2nd char	ASCII
#3	VIN – 3rd char	ASCII
#4	VIN – 4th char	ASCII
#5	VIN – 5th char	ASCII
#6	VIN - 6th char	ASCII
#7	VIN – 7th char	ASCII
#8	VIN – 8th char	ASCII
#9	VIN – 9th char	ASCII
#10	VIN - 10th char	ASCII
#11	VIN - 11th char	ASCII
#12	VIN - 12th char	ASCII
#13	VIN – 13th char	ASCII
#14	VIN – 14th char	ASCII
#15	VIN – 15th char	ASCII
#16	VIN – 16th char	ASCII
#17	VIN – 17th char	ASCII

4.3.1.8 DID $F192 systemSupplierECUHardwareNumber (Read)

DID Byte Index	Definition	Display
#1	DBCM ECU Base Model Part Number MSB	USN
#2	DBCM ECU Base Model Part Number 2nd MSB	USN
#3	DBCM ECU Base Model Part Number 2nd LSB	USN
#4	DBCM ECU Base Model Part Number LSB	USN

4.3.1.9 DID $F193 systemSupplierECUHardwareVersionNumber (Read)

DID Byte Index	Definition	Display
#1	DBCM ECU Base Model Part Number for Design Level Suffix MSB	ASCII
#2	DBCM ECU Base Model Part Number for Design Level Suffix LSB	ASCII

4.3.1.10 DID $F194 systemSupplierECUSoftwareNumber (Read)

DID Byte Index	Definition	Display
#1	DBCM ECU SoftwareNumber MSB	USN
#2	DBCM ECU SoftwareNumber 2nd MSB	USN
#3	DBCM ECU SoftwareNumber 2nd LSB	USN
#4	DBCM ECU SoftwareNumber LSB	USN

4.3.1.11 DID $F195 SystemSupplierECUSoftwareVersionNumberDataIdentifier(Read)

DID Byte Index	Definition	Display
#1	EBCM ECU Software Part Number MSB	ASCII
#2	EBCM ECU Software Part Number 2nd MSB	ASCII

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

DID Byte Index	Definition	Display
#3	EBCM ECU Software Part Number	ASCII
#4	EBCM ECU Software Part Number	ASCII
#5	EBCM ECU Software Part Number	ASCII
#6	EBCM ECU Software Part Number	ASCII
#7	EBCM ECU Software Part Number	ASCII
#8	EBCM ECU Software Part Number	ASCII
#9	EBCM ECU Software Part Number 2nd LSB	ASCII
#10	EBCM ECU Software Part Number LSB	ASCII

- Note: If spare bytes, please set 0 in LSB bytes

4.3.1.12 DID $F197 SystemName(Read)

DID Byte Index	Definition	Display
#1	41	ASCII
#2	42	ASCII
#3	53	ASCII
#4	20	ASCII
#5	20	ASCII
#6	20	ASCII
#7	20	ASCII
#8	20	ASCII
#9	20	ASCII
#10	20	ASCII

4.3.1.13 DID $F199 Date ECU Reprogrammed (Read/ Write*)

DID Byte Index	Definition	Display
#1	Year (MSB)	USN
#2	Year (LSB)	USN
#3	Month	USN
#4	Day	USN

4.3.1.14 DID $F19B CalibrationDateInformation (Read/Write)

DID Byte Index	Definition	Display
#1	Calibration Date – 1st char	ASCII
#2	Calibration Date – 2nd char	ASCII
#3	Calibration Date – 3rd char	ASCII
#4	Calibration Date – 4th char	ASCII
#5	Calibration Date – 5th char	ASCII
#6	Calibration Date - 6th char	ASCII
#7	Calibration Date – 7th char	ASCII
#8	Calibration Date – 8th char	ASCII
#9	Calibration Date – 9th char	ASCII
#10	Calibration Date - 10th char	ASCII
#11	Calibration Date - 11th char	ASCII
#12	Calibration Date - 12th char	ASCII
#13	Calibration Date – 13th char	ASCII
#14	Calibration Date – 14th char	ASCII
#15	Calibration Date – 15th char	ASCII
#16	Calibration Date – 16th char	ASCII

4.3.1.15 DID $F19D ECUInstallationDateDataIdentifier (Read/ Write)

DID Byte Index	Definition	Display

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

DID Byte Index	Definition	Display
#1	ECU Installation Date – 1st char	ASCII
#2	ECU Installation Date – 2nd char	ASCII
#3	ECU Installation Date – 3rd char	ASCII
#4	ECU Installation Date – 4th char	ASCII

4.3.1.16 DID $F1A0 OEMSpecific_2(Read/Write)

DID Byte Index	Definition	Display
#1	OEMSpecific_2 – 1st char	ASCII
#2	OEMSpecific_2 – 2nd char	ASCII
#3	OEMSpecific_2 – 3rd char	ASCII
#4	OEMSpecific_2 – 4th char	ASCII
#5	OEMSpecific_2 – 5th char	ASCII
#6	OEMSpecific_2 - 6th char	ASCII
#7	OEMSpecific_2 – 7th char	ASCII
#8	OEMSpecific_2 – 8th char	ASCII
#9	OEMSpecific_2 – 9th char	ASCII
#10	OEMSpecific_2 - 10th char	ASCII
#11	OEMSpecific_2 - 11th char	ASCII
#12	OEMSpecific_2 - 12th char	ASCII
#13	OEMSpecific_2 – 13th char	ASCII
#14	OEMSpecific_2 – 14th char	ASCII
#15	OEMSpecific_2 – 15th char	ASCII
#16	OEMSpecific_2 – 16th char	ASCII

4.3.1.17 DID $F1A1 OEMSpecific_3(Read/Write)

DID Byte Index	Definition	Display
#1	OEMSpecific_3 – 1st char	ASCII
#2	OEMSpecific_3 – 2nd char	ASCII
#3	OEMSpecific_3 – 3rd char	ASCII
#4	OEMSpecific_3 – 4th char	ASCII
#5	OEMSpecific_3 – 5th char	ASCII
#6	OEMSpecific_3 - 6th char	ASCII
#7	OEMSpecific_3 – 7th char	ASCII
#8	OEMSpecific_3 – 8th char	ASCII
#9	OEMSpecific_3 – 9th char	ASCII
#10	OEMSpecific_3 - 10th char	ASCII
#11	OEMSpecific_3 - 11th char	ASCII
#12	OEMSpecific_3 - 12th char	ASCII
#13	OEMSpecific_3 – 13th char	ASCII
#14	OEMSpecific_3 – 14th char	ASCII
#15	OEMSpecific_3 – 15th char	ASCII
#16	OEMSpecific_3 – 16th char	ASCII

4.3.1.18 DID $3003 SupplierEHCUTraceability (Read/Write)

DID Byte Index	Definition	Display
#1	DBCM EHCU Manufacture Code MSB	ASCII
#2	DBCM EHCU Manufacture Code MB	ASCII
#3	DBCM EHCU Manufacture Code LSB	ASCII
#4	DBCM EHCU Date Code MSB	ASCII
#5	DBCM EHCU Date Code MB	ASCII
#6	DBCM EHCU Date Code LSB	ASCII

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

DID Byte Index	Definition	Display
#7	DBCM EHCU Year Code	ASCII
#8	DBCM EHCU Broadcast Code MSB	ASCII
#9	DBCM EHCU Broadcast Code 2nd MSB	ASCII
#10	DBCM EHCU Broadcast Code 2nd LSB	ASCII
#11	DBCM EHCU Broadcast Code LSB	ASCII

4.3.1.19 DID $3020 ManufacturersEnableCounter (Read/Write)

DID Byte Index	Definition	Display
#1	Manufacturing Enable Counter	USN

- Writable only if the Manufacturing Enable Counter is not equal to 0.

4.3.2 Supplier Defined Dynamic DataIdentifier

4.3.2.1 DID $3007 System Speed and Status Information

Record Value #	Bit #	Definition	Units	Range	Reporting Resolution
1 2		LF Wheel Speed - unfiltered	KPH	0-255	0.0078125
3 4		RF Wheel Speed – unfiltered	KPH	0-255	0.0078125
5 6		LR Wheel speed - unfiltered	KPH	0-255	0.0078125
7 8		RR Wheel Speed – unfiltered	KPH	0-255	0.0078125
9 10		Filtered Vehicle Speed	KPH	0-255	0.015625
11		Rough Road Parameter	G	0-1	0.00392
12	7	LF State of Wheel Speed Electrical Status.	State	N/A	0 =NORMAL 1= FAILED
	6	RF State of Wheel Speed Electrical Status.	State	N/A	0 =NORMAL 1= FAILED
	5	LR State of Wheel Speed Electrical Status.	State	N/A	0 =NORMAL 1= FAILED
	4	RR State of Wheel Speed Electrical Status.	State	N/A	0 =NORMAL 1= FAILED
	3	Unused / Reserved			
	2	AWD Switch State	State	N/A	1 = AWD Active
	1	ABS Active	State	N/A	1 = True
	0	ABS Brake Switch (based upon brake sensor only)	State	N/A	1 = True
13 14		System_Voltage	VOLTS	0-19	1/1024
15 16		Battery Voltage	VOLTS	0-19	1/1024
17		Unused	N/A	N/A	N/A

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Record Value #	Bit #	Definition	Units	Range	Reporting Resolution
18 19		System State Operation	N/A	0-12	num

4.3.2.2 DID $3008 Power, Motor and Solenoid Information

Record Value #	Bit #	Definition	Units	Range	Reporting Resolution
1	All	Pump Motor Current	Amps	0 to 128	1/64
2					
3	All	Unused / Reserved	N/A	N/A	N/A
4					
5	All	Battery Voltage	Volts	0-19	1/1024
6					
7	All	Pump Voltage Across Feedback	Volts	0-19	1/1024
8					
9	All	System Voltage	Volts	0-19	1/1024
10					
11	All	Motor Drive Temperature	Degrees	-40 to 215	1
12					
13	7	Right Rear Release ABS Solenoid Activated	N/A	N/A	0=OFF 1=ON
	6	Right Rear Hold ABS Solenoid Activated	N/A	N/A	0=OFF 1=ON
	5	Left Rear Release ABS Solenoid Activated	N/A	N/A	0=OFF 1=ON
	4	Left Rear Hold ABS Solenoid Activated	N/A	N/A	0=OFF 1=ON
	3	Right Front Release ABS Solenoid Activated	N/A	N/A	0=OFF 1=ON
	2	Right Front Hold ABS Solenoid Activated	N/A	N/A	0=OFF 1=ON
	1	Left Front Release ABS Solenoid Activated	N/A	N/A	0=OFF 1=ON
	0	Left Front Hold ABS Solenoid Activated	N/A	N/A	0=OFF 1=ON
14	7	Unused / Reserved	N/A	N/A	N/A
	6	Unused / Reserved	N/A	N/A	N/A
	5	Unused / Reserved	N/A	N/A	N/A
	4	Unused / Reserved	N/A	N/A	N/A
	3	Unused / Reserved	N/A	N/A	N/A
	2	Unused / Reserved	N/A	N/A	N/A
	1	Pump Enable Command	N/A	N/A	0=OFF 1=ON
	0	Unused / Reserved	N/A	N/A	N/A
15		Unused / Reserved	N/A	N/A	N/A
16					
17		Unused / Reserved	N/A	N/A	N/A

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Record Value #	Bit #	Definition	Units	Range	Reporting Resolution
18		Unused / Reserved	N/A	N/A	N/A

4.3.2.3 DID $3009 Current And History ECU MCU DTC Information

Record Value #	Definition	Units	Range	Reporting Resolution
1	DBCM Current MCU Fault SincePowerUp 1	N/A	N/A	1=set
2	DBCM Current MCU Fault SincePowerUp 2	N/A	N/A	1=set
3	DBCM Current MCU Fault SincePowerUp 3	N/A	N/A	1=set
4	DBCM Current MCU Fault SincePowerUp 4	N/A	N/A	1=set
5	DBCM Current MCU Fault SincePowerUp 5	N/A	N/A	1=set
6	DBCM Current MCU Fault SincePowerUp 6	N/A	N/A	1=set
7~9	Unused	N/A	N/A	N/A
10	DBCM Current ECU Fault SincePowerUp 1	N/A	N/A	1=set
11	DBCM Current ECU Fault SincePowerUp2	N/A	N/A	1=set
12	DBCM Current ECU Fault SincePowerUp 3	N/A	N/A	1=set
13~ 16	Unused	N/A	N/A	N/A
17	DBCM History MCU Faults 1	N/A	N/A	1=set
18	DBCM History MCU Faults 2	N/A	N/A	1=set
19	DBCM History MCU Faults 3	N/A	N/A	1=set
20	DBCM History MCU Faults 4	N/A	N/A	1=set
21	DBCM History MCU Faults 5	N/A	N/A	1=set
22	DBCM History MCU Faults 6	N/A	N/A	1=set
23 ~ 25	Unused	N/A	N/A	N/A
26	DBCM History ECU Faults 1	N/A	N/A	1=set
27	DBCM History ECU Faults 2	N/A	N/A	1=set
28	DBCM History ECU Faults 3	N/A	N/A	1=set
29	Unused	N/A	N/A	N/A

M = 9, N = 7

4.3.2.4 DID $300A Last Fault Diagnostic Information

Record Value #	Bit #	Definition	Units	Range	Reporting Resolution
1		Last Fault Code Set.	N/A	N/A	BCD

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Record Value #	Bit #	Definition	Units	Range	Reporting Resolution
2 3		Same Format as service $19			
4	7	Unused / Reserved	N/A	N/A	N/A
	6	DRP Active When Last DTC Occurred	N/A	N/A	1=TRUE
	5	CBD Active When Last DTC Last Occurred	N/A	N/A	1=TRUE
	4	Front ABS Active When Last DTC Last Occurred	N/A	N/A	1=TRUE
	3	Rear ABS Active When Last DTC Last Occurred	N/A	N/A	1=TRUE
	2	Brake Switch On When Last DTC Last Occurred	N/A	N/A	1=TRUE
	1	Unused / Reserved	N/A	N/A	N/A
	0	Engine Running When Last DTC Last Occurred	N/A	N/A	1=TRUE
5	7	Unused / Reserved	N/A	N/A	N/A
	6	Unused / Reserved	N/A	N/A	N/A
	5	Unused / Reserved	N/A	N/A	N/A
	4	ABS Rough Road Active When Last DTC Last Occurred	N/A	N/A	1=TRUE
	3	Unused / Reserved	N/A	N/A	N/A
	2	Unused / Reserved	N/A	N/A	N/A
	1	Unused / Reserved	N/A	N/A	N/A
	0	Unused / Reserved	N/A	N/A	N/A
6		Vehicle Speed when Last Fault Code Last Occurred	kph	N/A	1
7		Vehicle Accelaration when Last Fault Code Last Occurred	g	-2 to +2	1/64
8		Ignition Cycles Since Last Fault Code last Occurred	count	N/A	1
9		Ignition Cycles Since Last Fault Code first occurred	count	0~255	1
10		Number of Occurrences of Last Fault Code	count	0~255	1
11		Left Front Wheel Speed when Last Fault Code Last Occurred	kph	0~255	1
12		Right Front Wheel Speed when Last Fault Code Last Occurred	kph	0~255	1
13		Left Rear Wheel Speed when Last Fault Code Last Occurred	kph	0~255	1
14		Right Rear Wheel Speed when Last Fault Code Last Occurred	kph	0~255	1
15 16		The accumulated driving time when the last fault last happened	seconds	0~65535	1
17		Unused/Reserved	N/A	N/A	N/A
18		Unused/Reserved	N/A	N/A	N/A

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

4.3.2.5 DID $300B Second Last Fault Diagnostic Information

Record Value #	Bit #	Definition	Units	Range	Reporting Resolution
1 2 3		Second Last Fault Code Set. Same Format as service $19	N/A	N/A	BCD
4	7	TCS Active When Second Last DTC Last Occurred	N/A	N/A	1=TRUE
	6	DRP Active When Second Last DTC Occurred	N/A	N/A	1=TRUE
	5	CBD Active When Second Last DTC Last Occurred	N/A	N/A	1=TRUE
	4	Front ABS Active When Second Last DTC Last Occurred	N/A	N/A	1=TRUE
	3	Rear ABS Active When Second Last DTC Last Occurred	N/A	N/A	1=TRUE
	2	Front Brake Switch On When Second Last DTC Last Occurred	N/A	N/A	1=TRUE
	1	Rear Brake Switch On When Second Last DTC Last Occurred	N/A	N/A	1=TRUE
	0	Engine Running When Second Last DTC Last Occurred	N/A	N/A	1=TRUE
5	7	Not Used	N/A	N/A	N/A
	6	Not Used	N/A	N/A	N/A
	5	Not Used	N/A	N/A	N/A
	4	ABS Rough Road Active When Second Last DTC Last Occurred	N/A	N/A	1=TRUE
	3	Steer Angle Centered When Second Last DTC Last Occurred	N/A	N/A	1=TRUE
	2	VSE Enabled When Second Last DTC Last Occurred	N/A	N/A	1=TRUE
	1	TCS/ETS Enabled When Second Last DTC Last Occurred	N/A	N/A	1=TRUE
	0	VSE Active When Second Last DTC Last Occurred	N/A	N/A	1=TRUE
6		Vehicle Speed when Second Last Fault Code Last Occurred	kph	N/A	1
7		Vehicle Accelaration when Second Last Fault Code Last Occurred	G	-2 to +2	1/64
8		Ignition Cycles Since Second Last Fault Code last Occurred	count	N/A	1/64
9		Ignition Cycles Since Second Last Fault Code first occurred	count	0~255	1
10		Number of Occurrences of Second Last Fault Code	count	0~255	1
11		Left Front Wheel Speed when Second Last Fault Code Last Occurred	kph	0~255	1

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Record Value #	Bit #	Definition	Units	Range	Reporting Resolution
12		Right Front Wheel Speed when Second Last Fault Code Last Occurred	kph	0~255	1
13		Left Rear Wheel Speed when Second Last Fault Code Last Occurred	kph	0~255	1
14		Right Rear Wheel Speed when Second Last Fault Code Last Occurred	kph	0~255	1
15 16		The accumulated driving time when the Second Last fault last happened	seconds	0 – 65535	1
17		Reserved			
18		Reserved			

4.3.2.6 DID $300C Third Last Fault Diagnostic Information

Record Value #	Bit #	Definition	Units	Range	Reporting Resolution
1 2 3		Third Last Fault Code Set. Same Format as service $19	N/A	N/A	BCD
4	7	TCS Active When Third Last DTC Last Occurred	N/A	N/A	1=TRUE
	6	DRP Active When Third Last DTC Occurred	N/A	N/A	1=TRUE
	5	CBD Active When Third Last DTC Last Occurred	N/A	N/A	1=TRUE
	4	Front ABS Active When Third Last DTC Last Occurred	N/A	N/A	1=TRUE
	3	Rear ABS Active When Third Last DTC Last Occurred	N/A	N/A	1=TRUE
	2	Front Brake Switch On When Third Last DTC Last Occurred	N/A	N/A	1=TRUE
	1	Rear Brake Switch On When Third Last DTC Last Occurred	N/A	N/A	1=TRUE
	0	Engine Running When Third Last DTC Last Occurred	N/A	N/A	1=TRUE
5	7	Not Used	N/A	N/A	N/A
	6	Not Used	N/A	N/A	N/A
	5	Not Used	N/A	N/A	N/A
	4	ABS Rough Road Active When Third Last DTC Last Occurred	N/A	N/A	1=TRUE
	3	Steer Angle Centered When Third Last DTC Last Occurred	N/A	N/A	1=TRUE
	2	VSE Enabled When Third Last DTC Last Occurred	N/A	N/A	1=TRUE
	1	TCS/ETS Enabled When Third Last DTC Last Occurred	N/A	N/A	1=TRUE

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Record Value #	Bit #	Definition	Units	Range	Reporting Resolution
	0	VSE Active When Third Last DTC Last Occurred	N/A	N/A	1=TRUE
6		Vehicle Speed When Third Last Fault Code Last Occurred	kph	N/A	1
7		Vehicle Accelaration When Third Last Fault Code Last Occurred	g	-2 to +2	1/64
8		Ignition Cycles Since Third Last Fault Code last Occurred	count	N/A	1/64
9		Ignition Cycles Since Third Last Fault Code first occurred	count	0~255	1
10		Number of Occurrences of Third Last Fault Code	count	0~255	1
11		Left Front Wheel Speed When Third Last Fault Code Last Occurred	kph	0~255	1
12		Right Front Wheel Speed When Third Last Fault Code Last Occurred	kph	0~255	1
13		Left Rear Wheel Speed When Third Last Fault Code Last Occurred	kph	0~255	1
14		Right Rear Wheel Speed When Third Last Fault Code Last Occurred	kph	0~255	1
15 16		The accumulated driving time when the Third Last fault last happened	seconds	0 – 65535	1
17		Reserved			
18		Reserved			

4.3.2.7 DID $3017 Fourth Last Fault Diagnostic Information

Record Value #	Bit #	Definition	Units	Range	Reporting Resolution
1 2 3		4th Last Fault Code Set. Same Format as service $19	N/A	N/A	BCD
4	7	TCS Active When 4th Last DTC Last Occurred	N/A	N/A	1=TRUE
	6	DRP Active When 4th Last DTC Occurred	N/A	N/A	1=TRUE
	5	CBD Active When 4th Last DTC Last Occurred	N/A	N/A	1=TRUE

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Record Value #	Bit #	Definition	Units	Range	Reporting Resolution
	4	Front ABS Active When 4th Last DTC Last Occurred	N/A	N/A	1=TRUE
	3	Rear ABS Active When 4th Last DTC Last Occurred	N/A	N/A	1=TRUE
	2	Front Brake Switch On When 4th Last DTC Last Occurred	N/A	N/A	1=TRUE
	1	Rear Brake Switch On When 4th Last DTC Last Occurred	N/A	N/A	1=TRUE
	0	Engine Running When 4th Last DTC Last Occurred	N/A	N/A	1=TRUE
5	7	Not Used	N/A	N/A	N/A
	6	Not Used	N/A	N/A	N/A
	5	Not Used	N/A	N/A	N/A
	4	ABS Rough Road Active When 4th Last DTC Last Occurred	N/A	N/A	1=TRUE
	3	Steer Angle Centered When 4th Last DTC Last Occurred	N/A	N/A	1=TRUE
	2	VSE Enabled When 4th Last DTC Last Occurred	N/A	N/A	1=TRUE
	1	TCS/ETS Enabled When 4th Last DTC Last Occurred	N/A	N/A	1=TRUE
	0	VSE Active When 4th Last DTC Last Occurred	N/A	N/A	1=TRUE
6		Vehicle Speed When 4th Last Fault Code Last Occurred	kph	N/A	1
7		Vehicle Accelaration When 4th Last Fault Code Last Occurred	g	-2 to +2	1/64
8		Ignition Cycles Since 4th Last Fault Code last Occurred	count	N/A	1/64
9		Ignition Cycles Since 4th Last Fault Code first occurred	count	0~255	1
10		Number of Occurrences of 4th Last Fault Code	count	0~255	1
11		Left Front Wheel Speed When 4th Last Fault Code Last Occurred	kph	0~255	1
12		Right Front Wheel Speed When 4th Last Fault Code Last Occurred	kph	0~255	1
13		Left Rear Wheel Speed When 4th Last Fault Code Last Occurred	kph	0~255	1
14		Right Rear Wheel Speed When 4th Last Fault Code Last Occurred	kph	0~255	1
15		The accumulated driving time when the 4th Last fault last happened	seconds	0 – 65535	1
16					
17		Reserved			
18		Reserved			

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

4.3.2.8 DID $3018 Fifth Last Fault Diagnostic Information

Record Value #	Bit #	Definition	Units	Range	Reporting Resolution
1 2 3		Fifth Last Fault Code Set. Same Format as service $19	N/A	N/A	BCD
4	7	TCS Active When Fifth Last DTC Last Occurred	N/A	N/A	1=TRUE
	6	DRP Active When Fifth Last DTC Occurred	N/A	N/A	1=TRUE
	5	CBD Active When Fifth Last DTC Last Occurred	N/A	N/A	1=TRUE
	4	Front ABS Active When Fifth Last DTC Last Occurred	N/A	N/A	1=TRUE
	3	Rear ABS Active When Fifth Last DTC Last Occurred	N/A	N/A	1=TRUE
	2	Front Brake Switch On When Fifth Last DTC Last Occurred	N/A	N/A	1=TRUE
	1	Rear Brake Switch On When Fifth Last DTC Last Occurred	N/A	N/A	1=TRUE
	0	Engine Running When Fifth Last DTC Last Occurred	N/A	N/A	1=TRUE
5	7	Not Used	N/A	N/A	N/A
	6	Not Used	N/A	N/A	N/A
	5	Not Used	N/A	N/A	N/A
	4	ABS Rough Road Active When Fifth Last DTC Last Occurred	N/A	N/A	1=TRUE
	3	Steer Angle Centered When Fifth Last DTC Last Occurred	N/A	N/A	1=TRUE
	2	VSE Enabled When Fifth Last DTC Last Occurred	N/A	N/A	1=TRUE
	1	TCS/ETS Enabled When Fifth Last DTC Last Occurred	N/A	N/A	1=TRUE
	0	VSE Active When Fifth Last DTC Last Occurred	N/A	N/A	1=TRUE
6		Vehicle Speed When Fifth Last Fault Code Last Occurred	kph	N/A	1
7		Vehicle Accelaration When Fifth Last Fault Code Last Occurred	g	-2 to +2	1/64
8		Ignition Cycles Since Fifth Last Fault Code last Occurred	count	N/A	1/64
9		Ignition Cycles Since Fifth Last Fault Code first occurred	count	0~255	1
10		Number of Occurrences of Fifth Last Fault Code	count	0~255	1
11		Left Front Wheel Speed When Fifth Last Fault Code Last Occurred	kph	0~255	1

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Record Value #	Bit #	Definition	Units	Range	Reporting Resolution
12		Right Front Wheel Speed When Fifth Last Fault Code Last Occurred	kph	0~255	1
13		Left Rear Wheel Speed When Fifth Last Fault Code Last Occurred	kph	0~255	1
14		Right Rear Wheel Speed When Fifth Last Fault Code Last Occurred	kph	0~255	1
15		The accumulated driving time when the Fifth Last fault last happened	seconds	0 – 65535	1
16					
17		Reserved			
18		Reserved			

4.3.2.9 DID $300D CurrentDiagnosticDTCInformation

Record Value #	Definition	Units	Range	Reporting Resolution
1	DBCM TestFailedThisOperationCycle 18 to 11	N/A	N/A	1= Set, 0=Not Set
2	DBCM TestFailedThisOperationCycle 28 to 21	N/A	N/A	1= Set, 0=Not Set
3	DBCM TestFailedThisOperationCycle 38 to 31	N/A	N/A	1= Set, 0=Not Set
4	DBCM TestFailedThisOperationCycle 48 to 41	N/A	N/A	1= Set, 0=Not Set
5	DBCM TestFailedThisOperationCycle 58 to 51	N/A	N/A	1= Set, 0=Not Set
6	DBCM TestFailedThisOperationCycle 68 to 61	N/A	N/A	1= Set, 0=Not Set
7	DBCM TestFailedThisOperationCycle 78 to 71	N/A	N/A	1= Set, 0=Not Set
8	DBCM TestFailedThisOperationCycle 88 to 81	N/A	N/A	1= Set, 0=Not Set
9	DBCM TestFailedThisOperationCycle 98 to 91	N/A	N/A	1= Set, 0=Not Set
10	DBCM TestFailedThisOperationCycle 108 to 101	N/A	N/A	1= Set, 0=Not Set
11	DBCM TestFailedThisOperationCycle 118 to 111	N/A	N/A	1= Set, 0=Not Set
12	DBCM TestFailedThisOperationCycle 128 to 121	N/A	N/A	1= Set, 0=Not Set
13	DBCM TestFailedThisOperationCycle 138 to 131	N/A	N/A	1= Set, 0=Not Set
14	DBCM TestFailedThisOperationCycle 148 to 141	N/A	N/A	1= Set, 0=Not Set
15	DBCM TestFailedThisOperationCycle 158 to 151	N/A	N/A	1= Set, 0=Not Set
16	DBCM TestFailedThisOperationCycle 168 to 161	N/A	N/A	1= Set, 0=Not Set
17	DBCM TestFailedThisOperationCycle 178 to 171	N/A	N/A	1= Set, 0=Not Set
18	DBCM TestFailedThisOperationCycle 188 to 181	N/A	N/A	1= Set, 0=Not Set
19	DBCM TestFailedThisOperationCycle 198 to 191	N/A	N/A	1= Set, 0=Not Set
20	DBCM TestFailedThisOperationCycle 208 to 201	N/A	N/A	1= Set, 0=Not Set
21	DBCM TestFailedThisOperationCycle 218 to 211	N/A	N/A	1= Set, 0=Not Set
22	DBCM CurrentDTCSincePowerUp 228 to 221	N/A	N/A	1=set

4.3.2.10 DID $300E HistoryDiagnosticDTCInformation

Record Value #	Definition	Units	Range	Reporting Resolution
1	DBCM HistoryDTC 18 to 11	N/A	N/A	1= Set, 0=Not Set
2	DBCM HistoryDTC 28 to 21	N/A	N/A	1= Set, 0=Not Set
3	DBCM HistoryDTC 38 to 31	N/A	N/A	1= Set, 0=Not Set
4	DBCM HistoryDTC 48 to 41	N/A	N/A	1= Set, 0=Not Set
5	DBCM HistoryDTC 58 to 51	N/A	N/A	1= Set, 0=Not Set

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Record Value #	Definition	Units	Range	Reporting Resolution
6	DBCM HistoryDTC 68 to 61	N/A	N/A	1= Set, 0=Not Set
7	DBCM HistoryDTC 78 to 71	N/A	N/A	1= Set, 0=Not Set
8	DBCM HistoryDTC 88 to 81	N/A	N/A	1= Set, 0=Not Set
9	DBCM HistoryDTC 98 to 91	N/A	N/A	1= Set, 0=Not Set
10	DBCM HistoryDTC 108 to101	N/A	N/A	1= Set, 0=Not Set
11	DBCM HistoryDTC 118 to 111	N/A	N/A	1= Set, 0=Not Set
12	DBCM HistoryDTC 128 to121	N/A	N/A	1= Set, 0=Not Set
13	DBCM HistoryDTC 138 to 131	N/A	N/A	1= Set, 0=Not Set
14	DBCM HistoryDTC 148 to 141	N/A	N/A	1= Set, 0=Not Set
15	DBCM HistoryDTC 158 to 151	N/A	N/A	1= Set, 0=Not Set
16	DBCM HistoryDTC 168 to 161	N/A	N/A	1= Set, 0=Not Set
17	DBCM HistoryDTC 178 to 171	N/A	N/A	1= Set, 0=Not Set
18	DBCM HistoryDTC 188 to 181	N/A	N/A	1= Set, 0=Not Set
19	DBCM HistoryDTC 198 to 191	N/A	N/A	1= Set, 0=Not Set
20	DBCM HistoryDTC 208 to 201	N/A	N/A	1= Set, 0=Not Set
21	DBCM HistoryDTC 218 to 211	N/A	N/A	1= Set, 0=Not Set
22	DBCM HistoryDTC 228 to 221	N/A	N/A	1=set

4.3.3 DID $3100 ~ $311F DeviceControl

4.3.3.1 DataIdentifier #1 Definition

Data Byte #	Bit #	Definition	Units	Range	Command Resolution
1	all	$31			
2	7~5	0	N/A	N/A	N/A
	4	Unused / Reserved	N/A	N/A	N/A
	3	Enable Solenoid PWM Control Command	N/A	N/A	Enable = 1, Disable = 0
	2	Enable solenoid ON-Off Control Command	N/A	N/A	Enable = 1, Disable = 0
	1	Enable Pump Motor Control Command	N/A	N/A	Enable = 1, Disable = 0
	0	Enable system Relay Control Command	N/A	N/A	Enable = 1, Disable = 0

4.3.3.2 Control State bytes definition

Control State byte	Bit #	Definition	Units	Range	Command Resolution
#1	7	Right Rear Release Solenoid On-Off Control Command	N/A	N/A	On = 1,Off= 0

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL

Control State byte	Bit #	Definition	Units	Range	Command Resolution
	6	Right Rear Apply Solenoid On-Off Control Command	N/A	N/A	On = 1,Off= 0
	5	Left Rear Release Solenoid On-Off Control Command	N/A	N/A	On = 1, Off= 0
	4	Left Rear Apply Solenoid On-Off Control Command	N/A	N/A	On = 1, Off= 0
	3	Right Front Release Solenoid On-Off Control Command	N/A	N/A	On = 1, Off= 0
	2	Right Front Apply Solenoid On-Off Control Command	N/A	N/A	On = 1, Off= 0
	1	Left Front Release Solenoid On-Off Control Command	N/A	N/A	On = 1, Off= 0
	0	Left Front Apply Solenoid On-Off Control Command	N/A	N/A	On = 1, Off= 0
#2	7	Unused / Reserved	N/A	N/A	N/A
	6	Unused / Reserved	N/A	N/A	N/A
	5	Unused / Reserved	N/A	N/A	N/A
	4	Unused / Reserved	N/A	N/A	N/A
	3	Unused / Reserved	N/A	N/A	N/A
	2	Unused / Reserved	N/A	N/A	N/A
	1	Pump Motor On/Off Control Command	N/A	N/A	On = 1, Off= 0
	0	System Relay Control Command	N/A	N/A	On = 1, Off= 0

Table 4.3-2 DID $3100-$31FF System Output Control

5 Revision History

Revision	Date	Description of Change
000	9/21/2009	Initial Release for DBC78
1.1	11/02/2009	Removed all “Delphi” References 2.1.2.1 Removed specific references to CAN diagnostic identiferes and made description generic for all customers 4.3.1. Removed documentation of DIDs containing EHCU sub-assembly information. ($F18C, $F192, $F193, $F194, $3001, $3002, $3004, $3005, $3006)
1.2	6/03/10	Corrected the Bleed example Device Control messages to enable motor and relay. Hex 4 to hex 3 for byte 6. Section 3.5.1.3.
1.2	6/03/10	6.4.1.1 scr 4624. Add more info to DID System wheel Speed Information 1. Add the following info to byte 12 the existing DID $3007: - AWD_Active - Vehicle_ABS_Active - Brake Switch Status 2. Append a byte which will have the system state information to the existing $3007 DID.
1.3	6/10/10	Corrected the version mistake on V1.2 which stated 1.1. Unreleased at 1.3
1.3	10-13-10	Added additional DID definitions and Security Service ID $27

DO NOT DISTRIBUTE OR COPY WITHOUT BWI APPROVAL